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A new test of factor model for asset returns: based on pleiotropy model

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The validity of factor model for asset returns relies on testing the null hypothesis of whether the intercepts are jointly indistinguishable from zero, which, however, is often rejected, resulting in imperfect comparison among competing asset pricing models. In this paper, we consider a new test based on the pleiotropy model, through which we could determine how many intercepts in the time-series regressions in portfolio-level are statistically non-zero and thus provide a more direct way to compare the validity of competing factor models. A sequential testing procedure regarding the estimated parameters under unconstrained and constrained conditions is proposed, based on some likelihood ratio test statistics. Simulation studies suggest that our proposed test statistics have better performance than the influential *GRS* test. Finally, we apply the proposed method to real data of portfolio returns with Fama-French five factor models, and demonstrate that it provides important and irreplaceable information for determining the efficacy of different asset pricing models.

Keywords: Pleiotropy model; Factor model; Asset returns; Sequential test

JEL Classification: C12, C13, G12

1. Introduction

Asset pricing has long been considered as a fundamental research question in financial economics. Beginning with the capital asset pricing model (CAPM) of Sharpe (1964) and Lintner (1965), the asset pricing literature has attempted to understand the determinants of risk premia on financial assets. Sharpe (1964) and Lintner (1965) argue that risk premium should depend on market beta or other measure(s) of systematic risk. In addition to the model of market excess return, perhaps the most influential model is the three-factor model of Fama and French (1993), which further includes a book-to-market or 'value' factor, HML (high-low) and a size factor, SMB (small-big). Fama and French (1993) show that this model captures most of the cross-sectional variation in average returns of 25 portfolios sorted by size and value.

Beyond Fama and French (1993), there is a substantial body of ad-hoc factor models indicating that several cross-sectional return predictability patterns are not subsumed by value and size. The empirical evidence accumulated since Fama and French (1996) suggests that there are other important sources

of cross-sectional variation in expected returns unrelated to the Fama-French three factors, including the Hou *et al.* (2015) four-factor model, the Fama and French (2015) five-factor model, the Stambaugh and Yuan (2017) four-factor model, the Fama and French (2018) six-factor models, the Barillas and Shanken (2018) six-factor model, the Daniel *et al.* (2020) three-factor model, and the Hou *et al.* (2020) five-factor model. Moreover, there are also many influential studies comparing the new factor models on both empirical and conceptual grounds, see Hou *et al.* (2015, 2019, 2021) and Hou and van Dijk (2019). Besides ad-hoc factor models, studies have also moved to developing asset pricing models that could test risk-based rational expectations explanations and sentiment explanations, such as ICAPM and long-run risk models, see Nagel (2013) for detailed summary. Moreover, Harvey *et al.* (2016) proposed a new framework that allowed for multiple tests and derived recommended statistical significance levels for research in asset pricing. Chordia *et al.* (2019) and Pukthuanthong *et al.* (2019) have proposed a protocol for identifying genuine risk factors based on individual stock level.

To test the validity of those factor models, research has built on Jensen (1968) and Black *et al.* (1972) to examine

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the intercepts in time-series regressions of excess test portfolio returns on market excess returns. Given that the market portfolio is efficient, the intercepts or ‘alphas’ of asset pricing models should be zero. A joint F-test of this hypothesis is later developed by Gibbons *et al.* (1989), or *GRS* statistic. However, as mentioned by Barillas and Shanken (2018), these tests often suggest ‘rejection’ of the implied restrictions, even in the Fama-French five-factor model, see Fama and French (2015), especially when the data sets are large (Fama and French 2016). To alleviate this problem, Fama and French (2015) compare the magnitude of *GRS* statistics for average returns on portfolios formed in different ways and claim that they only want to identify the the best (but imperfect) model. Feng *et al.* (2020) proposed a model selection method to systematically evaluate the contribution of asset pricing of any new factors. Indeed, a relatively large p -value may say more about imprecision in estimating a particular model’s alphas than the adequacy of that model. But statistically speaking, a lower p -value does not sufficiently indicate a good fit of a model with real data. Later, a large and established literature tries to compare misspecified asset pricing models, both in the general case of estimating SDFs and using tradable and nontradable factors (e.g. based on Hansen-Jagannathan distance, Sharpe ratio, or similar criteria), as well as doing model comparison and evaluation based on only tradable factors [e.g. the spanning test of Barillas and Shanken (2018), and its Siddhartha *et al.* (2020) correction]. Thus the existing methods mainly consider the problem whether all the intercepts are zero or not. But these tests could not identify how many and which intercepts are zero. In this paper, motivated by the pleiotropy model, that is, the sequential tests (Schaid *et al.* 2016, 2019) for identifying how many the traits affected by genes, we consider a new test with an aim to determine how many intercepts in the time-series regressions in portfolio-level are statistically non-zero, in which way we supplement this strand of research by finding a more direct way to compare competing asset pricing models. We extend the works of Schaid *et al.* (2016, 2019) by relaxing the restrictions on the moments of random variables and allowing estimating models with known likelihood functions. We demonstrate that our proposed method provides important and irreplaceable information, compared to the *GRS* test, for determining the efficacy of asset pricing models.

The concept of pleiotropy originates from genetics, where it describes a situation in which a single gene affects multiple traits. By detecting the extent to which a gene influences various traits and understanding the underlying causes, researchers can gain deeper insights into the gene’s biological functions. Applying the idea of pleiotropy to asset pricing, we can draw a parallel by considering the intercepts in portfolio-level regressions (often referred to as abnormal returns) as analogous to the ‘single gene’. In a well-specified linear asset pricing model, all variations in expected returns should be captured by exposures to risk factors, resulting in zero intercepts for all securities and portfolios. However, if some portfolios exhibit abnormal returns (non-zero intercepts), it indicates that the model fails to fully capture the return variation. Thus, testing the adequacy of an asset pricing model is akin to detecting the degree of pleiotropy in the ‘intercept gene’. Specifically, it involves evaluating how many of the

intercepts in the time-series regressions across portfolios are statistically different from zero, which parallels assessing how many traits a gene can influence in genetic studies.

In this paper, we extend the pleiotropy model of Schaid *et al.* (2016, 2019), and propose a sequential testing procedure for the estimated parameters under both unconstrained and constrained conditions, using likelihood ratio test statistics. Our simulation studies show that the proposed test statistics outperform the widely used *GRS* test in asset pricing. We then apply the method to real portfolio return data with Fama-French five factor models, demonstrating its ability to provide crucial and unique insights into the effectiveness of various asset pricing models. Based on these findings, we recommend using our extended pleiotropy model as a tool for evaluating the efficacy of asset pricing models.

Our paper contributes to literature on asset pricing by providing a more direct and potentially more useful way to test the validity of factor models and compare the efficacy among them. As stated above, existing literature mainly relies on *GRS* statistic, which often suggest ‘rejection’ of the implied restrictions. Our method is empirically more efficient, as demonstrated later in the extensive simulation, in comparing different factor models and thus has better performance than *GRS*.

Our research also contributes to literature both by developing a better estimating and testing procedure for pleiotropy model, and by providing a pleiotropy test for generalized linear model. This is conducted by firstly estimating the parameters under the null hypothesis, which is equivalent to obtain the maximum likelihood function under different constraints and then proposing a combined test statistic based on some likelihood ratio test statistics between the unconstrained and constrained conditions.

The rest of the paper is organized as follows. We begin in Section 2 by introducing a generalized linear model for our approach with a description of the estimation procedure for the model. In Section 3, we describe how to divide our hypothesis test to construct test statistics regarding unconstrained and constrained conditions, investigate the asymptotic properties of all estimators and test statistics, and provide a practical procedure to determine important parameters. Simulation studies are conducted in Section 4 to show how accurate the proposed estimators are and how powerful the proposed hypothesis test is. In Section 5, we apply the proposed modeling, estimation procedure and hypothesis test to the real data set, and some interesting findings will be presented. Finally, Section 6 concludes with some discussion. Proofs of the theoretical results and testing procedure for VAR(1) model are given in the Appendix.

2. Model and estimation

2.1. Model and notations

Suppose that $\mathbf{y} = (y_1, \dots, y_p)^\top$ is a p -dimensional random vector of interest[†] (such as portfolio return in the empirical analysis) and $\mathbf{x}$ is the covariate (factor) that may affect $\mathbf{y}$.

[†] The notation $^\top$ denotes the transpose of a vector or matrix.

Suppose that $\mathbf{y}$ has the following conditional density function

$$f_{\mathbf{y}}(\mathbf{y} | \mathbf{x}) = f(\mathbf{y} | \mathbf{x}^\top \boldsymbol{\beta}, \boldsymbol{\lambda}) \quad (1)$$

where $\boldsymbol{\lambda}$ is the nuisance parameter. $\mathbf{x} = (x_1, \dots, x_q)^\top$ is a q -dimensional random vector (the first component of $\mathbf{x}$ is 1 and is used to estimate the intercept β_1 , which is equivalent to abnormal return, alpha, in asset pricing model) and the coefficients $\boldsymbol{\beta}$ is a $q \times p$ -dimensional matrix of interest, which characterize the factor loadings. For notational simplicity, let $X = \mathbf{I}_p \otimes \mathbf{x}$, and $\boldsymbol{\beta} = \text{vec}(\boldsymbol{\beta})$, where $\mathbf{I}_p$ is a $p \times p$ identity matrix. Thus it holds that $\mathbf{x}^\top \boldsymbol{\beta} = X^\top \boldsymbol{\beta}$. In our setup, the random vector $\mathbf{y}$ can be continuous or discrete variable. For example, if the random vector $\mathbf{y}$ is discrete, the density function $f_{\mathbf{y}}(\cdot)$ may be the logistic or probit density function; if the random vector $\mathbf{y}$ is continuous like portfolio return, some classical continuous distributions can also be included in our framework. In the sequel, we present some examples:

- If the random vector $\mathbf{y}$ follows a multivariate normal distribution, then the model (1) can be written as

$$\begin{aligned} f(\mathbf{y} | X^\top \boldsymbol{\beta}, \Sigma) \\ = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \\ \times \left\{ -\frac{1}{2} (\mathbf{y} - X^\top \boldsymbol{\beta})^\top \Sigma^{-1} (\mathbf{y} - X^\top \boldsymbol{\beta}) \right\}. \end{aligned}$$

Here $\boldsymbol{\lambda} = \Sigma$. The maximum likelihood estimation of $\boldsymbol{\beta}$ is also equivalent to the least square estimation, which is thoroughly discussed in Schaid *et al.* (2016).

- If the random vector $\mathbf{y}$ follows a logistic model, thus $\mathbf{y}$ is a discrete random vector. For simplicity, we consider a marginal distribution of $\mathbf{y}$, that could be

$$\begin{aligned} f\{y_j | \mathbf{x}^\top \boldsymbol{\beta}_{(j)}\} &= \left[\frac{\exp\{\mathbf{x}^\top \boldsymbol{\beta}_{(j)}\}}{1 + \exp\{\mathbf{x}^\top \boldsymbol{\beta}_{(j)}\}} \right]^{y_j} \\ &\times \left[\frac{1}{1 + \exp\{\mathbf{x}^\top \boldsymbol{\beta}_{(j)}\}} \right]^{1-y_j}, \end{aligned}$$

where y_j is the j th component of $\mathbf{y}$ and $\boldsymbol{\beta}_{(j)}$ is the j th column vector of $\boldsymbol{\beta}$.

- If the random vector $\mathbf{y}$ follows a Box-Cox model, then we have

$$\begin{aligned} f(\mathbf{y} | X^\top \boldsymbol{\beta}, \boldsymbol{\lambda}, \Sigma) \\ = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} \{G(\mathbf{y}; \boldsymbol{\lambda}) - X^\top \boldsymbol{\beta}\}^\top \right. \\ \left. \times \Sigma^{-1} \{G(\mathbf{y}; \boldsymbol{\lambda}) - X^\top \boldsymbol{\beta}\} \right], \end{aligned}$$

where $G(\mathbf{y}; \boldsymbol{\lambda}) = \{G_1(y_1; \lambda_1), \dots, G_p(y_p; \lambda_p)\}^\top$ is a box-cox transformation function with $G_j(y_j; \lambda_j) = (y_j^{\lambda_j} - 1)/\lambda_j$ for $\lambda_j \neq 0$ and $\log(y_j)$ for $\lambda_j = 0$ for any $j = 1, \dots, p$. Moreover, $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_p)^\top$ and Σ are the nuisance parameters.

2.2. Estimation method

Let $\{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$ be a sequence of independent and identically distributed (i.i.d.) random samples with the sample distribution as $(\mathbf{x}, \mathbf{y})$ in model (1). Denote $\boldsymbol{\theta} = \{\boldsymbol{\beta}^\top, \text{vec}(\boldsymbol{\lambda})^\top\}^\top$. Since the density function of $\mathbf{y}$ is assumed to be known, a classical method for estimating parameters is available, that is, maximum likelihood estimation. Then under the model (1), the likelihood function of $\boldsymbol{\theta}$ takes the form

$$L_n(\boldsymbol{\theta}) = \prod_{i=1}^n f(\mathbf{y}_i | X_i^\top \boldsymbol{\beta}, \boldsymbol{\lambda}) = \prod_{i=1}^n f(\mathbf{y}_i | \mathbf{x}_i, \boldsymbol{\theta}). \quad (2)$$

The maximum likelihood estimator $\hat{\boldsymbol{\theta}}_n$ based on specific parametric distribution $f(\mathbf{y} | X^\top \boldsymbol{\beta}, \boldsymbol{\lambda})$ is obtained by maximizing the associated log-likelihood function $l_n(\boldsymbol{\theta}) = \log L_n(\boldsymbol{\theta})$, which is equivalent to solving the equation $\partial l_n(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} = \mathbf{0}$.

Write the true value of $\boldsymbol{\theta}$ as $\boldsymbol{\theta}_0$, and use $\xrightarrow{d}$ to denote convergence in distribution. The following lemma presents the asymptotic property of the maximum likelihood estimator $\hat{\boldsymbol{\theta}}_n$, which is also well-known in the theory of maximum likelihood estimation.

LEMMA 2.1 *Under Conditions (C1)–(C4) in Section 3.2, it holds that*

$$\begin{aligned} \sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \{I(\boldsymbol{\theta}_0)\}^{-1} \frac{\partial \log f(\mathbf{y}_i | \mathbf{x}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \\ &+ o_p(1) \xrightarrow{d} N(0, \{I(\boldsymbol{\theta}_0)\}^{-1}) \end{aligned} \quad (3)$$

as $n \rightarrow \infty$, where $I(\boldsymbol{\theta}_0)$ is the information matrix, that is,

$$I(\boldsymbol{\theta}_0) = \mathbb{E}_{\boldsymbol{\theta}_0} \left\{ -\frac{\partial^2 \log f(\mathbf{y}; \mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top} \right\}.$$

The asymptotic variance of $\hat{\boldsymbol{\theta}}_n$ can be consistently estimated by $\{I_n(\hat{\boldsymbol{\theta}}_n)\}^{-1}$ with $I_n(\boldsymbol{\theta}) = -n^{-1} \sum_{i=1}^n \partial^2 \log f(\mathbf{y}_i | \mathbf{x}_i, \boldsymbol{\theta}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top$.

In the following sections, we will conduct a hypothesis test about $\boldsymbol{\beta}$. Without loss of generality, we assume that λ is a scalar and $q = 1$ since the statistical inference below is easily extended to the general case $q > 1$. Then the parameter $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \lambda)^\top$ with $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$.

3. Hypothesis test

3.1. Test statistic construction

As described before, we would conduct a hypothesis test about pleiotropy in this section, which can be applied to determine how many intercepts in the time-series regressions in portfolio-level are statistically non-zero. Consider testing the null hypothesis of no pleiotropy, which means there is at most one portfolio that is affected by factor(s). This problem can be

formally stated as follows:

H_0 : Of the parameters $\beta_1, \dots, \beta_p$, there exists
at most one that is nonzero.

against the alternative hypothesis

H_1 : Otherwise.

Note that the null hypothesis is equivalent to testing whether one of the following $p + 1$ sub-hypotheses holds:

$$\begin{aligned} H_{00} : \beta_1 = \dots = \beta_p = 0; \\ H_{01} : \beta_1 \neq 0, \quad \beta_2 = \dots = \beta_p = 0. \\ \vdots \\ H_{0p} : \beta_p \neq 0, \quad \beta_1 = \dots = \beta_{p-1} = 0. \end{aligned}$$

It is easy to see that this test is quite different from the traditional hypothesis or multiple hypothesis test (MHT) since the latter performs a set of statistical tests simultaneously. But in our test, if H_0 holds, then there just exists one of H_{0k} 's holds, that is, only the k th portfolio is affected by factor(s). Let $V_0 = (I_p, 0_{p \times 1})$. Then $V_0\theta = \mathbf{0}$ is equivalent to $\beta_1 = \dots = \beta_p = 0$. When $k > 0$, we construct V_k by deleting the k th row from V_0 (to ignore β_k), which results in that $V_k\theta = (\beta_1, \dots, \beta_{k-1}, \beta_{k+1}, \dots, \beta_p)^\top$. To facilitate the statistical inference, we present these $p + 1$ hypotheses as suggested by Schaid *et al.* (2016, 2019) in the following form

- when $k = 0$, the sub-hypothesis H_{00} : $V_0\theta = \mathbf{0}$, and $V_j\theta = \mathbf{0}$ ($j = 1, \dots, p$);
- when $k > 0$, the sub-hypothesis H_{0k} : $V_k\theta = \mathbf{0}$, and $V_j\theta \neq \mathbf{0}$ ($j = 0, 1, \dots, k-1, k+1, \dots, p$).

Since the density function of $\mathbf{y}$ is assumed to be known, we could work within the framework of likelihood ratio test. For notational simplicity, let $\hat{\theta}_V = (\hat{\beta}_V^\top, \hat{\lambda}_V^\top)^\top$ be the estimator of θ that maximizes the log-likelihood function $l_n(\theta)$ under the constraint $V\theta = \mathbf{0}$, where V is a $s \times (p + 1)$ matrix with rank s ($s \leq p + 1$). Therefore under the null hypothesis H_{0k} : $V_k\theta = \mathbf{0}$, we construct a sequence of Log-likelihood Ratio Test (LRT) statistics as

$$T_{nk} = -2 \left\{ l_n(\hat{\theta}_{V_k}) - l_n(\hat{\theta}_n) \right\}, \quad (4)$$

where $k = 0, \dots, p$. In the following statements, we would present the theoretical properties of the LRT statistics, and discuss the detailed testing procedures based on the LRT statistics.

3.2. Theoretical properties

Similar to Schaid *et al.* (2016, 2019), we conclude some theorems and corollaries which follow from the properties of LRT statistics. To construct the theoretical properties of the proposed testing procedure, we need the following regularity conditions.

- (C1) The parameter space Θ of θ is compact and the true parameter θ_0 is an interior point of Θ .
- (C2) The true but unknown parameter value θ_0 is identified, i.e. $\theta_0 = \arg \max_{\theta \in \Theta} \mathbb{E}_{\theta_0} \log f(\mathbf{y}; \mathbf{x}, \theta)$ and $\mathbb{E}_{\theta_0} \log(\mathbf{y}; \mathbf{x}, \theta)$ exists.
- (C3) The log-likelihood function $l_n(\theta)$ is twice continuously differentiable in a neighborhood of θ_0 .
- (C4) The information matrix $I(\theta_0)$ exists and non-singular.

THEOREM 3.1 Assume that Conditions (C1)–(C4) hold. Let $A_n = nI_n(\hat{\theta}_n)$ and V be a $s \times d$ matrix of rank s ($s \leq d$), where $d = p + 1$ is the dimension of θ . Then the minimizer $\hat{\theta}_V$ of $(\theta - \hat{\theta}_n)^\top A_n (\theta - \hat{\theta}_n)$ under the constraint $V\theta = \mathbf{0}$ is $\hat{\theta}_V = \hat{\theta}_n - \tilde{\theta}_V$, where $\tilde{\theta}_V = A_n^{-1} V^\top [V A_n^{-1} V^\top]^{-1} V \hat{\theta}_n$. Furthermore, under the null hypothesis $V\theta_0 = \mathbf{0}$, it holds that

$$-2 \left\{ l_n(\hat{\theta}_V) - l_n(\hat{\theta}_n) \right\} = \tilde{\theta}_V^\top A_n \tilde{\theta}_V + o_p(1). \quad (5)$$

The proofs of Theorem 3.1 and other theorems are given in the Appendix. And it is easy to obtain the following important corollary based on Theorem 3.1, which can be viewed as the technical support for the construction of our test statistics.

COROLLARY 3.1 Assume that Conditions (C1)–(C4) hold. If $V\theta_0 = \mathbf{0}$, then $\tilde{\theta}_V A_n \tilde{\theta}_V \xrightarrow{d} \chi_s^2$ as $n \rightarrow \infty$. If $V\theta_0 \neq \mathbf{0}$, then $\tilde{\theta}_V A_n \tilde{\theta}_V = O_p(n)$.

REMARK 3.1 Corresponding to our test, we denote $\tilde{\theta}_{V_k}$ as the resulting estimator given in Theorem 3.1 under the constraints $V_k\theta_0 = \mathbf{0}$ ($k = 0, 1, \dots, p$). Then it follows from Theorem 3.1 that under sub-null hypothesis H_{0k} , the LRT statistic is

$$T_{nk} = \tilde{\theta}_{V_k}^\top A_n \tilde{\theta}_{V_k}, \quad (6)$$

where $k = 0, \dots, p$. Combining with Corollary 3.1, we can obtain the following theorem for the asymptotic properties of our proposed test statistics.

THEOREM 3.2 Assume that Conditions (C1)–(C4) hold.

- (1) If H_{00} holds, that is, there is no portfolio that is affected by factor(s), we have $T_{n0} \xrightarrow{d} \chi_p^2$ and $T_{nk} \xrightarrow{d} \chi_{p-1}^2$ for $k = 1, \dots, p$, as $n \rightarrow \infty$.
- (2) If only one H_{0k} ($k = 1, \dots, p$) holds, that is, there is at most one portfolio that is affected by factor(s), we have $T_{n0} = O_p(n)$ and $T_{\min} = \min_{k=1, \dots, p} T_{nk} \xrightarrow{d} \chi_{p-1}^2$ as $n \rightarrow \infty$.
- (3) If all H_{0k} 's ($k = 0, \dots, p$) do not hold, that is, there are more than one portfolios that are affected by factor(s), we have $T_{nk} = O_p(n)$ for $k = 0, \dots, p$.

Next, we will propose a testing procedure to test the null hypothesis.

3.3. Sequential testing

As Schaid *et al.* (2016, 2019) proposed, sequential testing provides a rigorous procedure to evaluate the number of portfolios associated with a factor. It is reasonable to accept the viewpoint that the more portfolios a factor associates with, the more important the factor is. And a sequential testing procedure is applicable to aid interpretation of how many and

which one of the portfolios are associated with a factor. With this purpose, we consider a null hypothesis, that is,

$H_0^{(m)}$: The number of nonzero components of the coefficient β is not greater than m .

H_0^m actually hypothesizes that there are no more than m portfolios that are affected by factor(s). Since the true value of m is unknown, we then consider a sequence of null hypotheses H_0^k , where $k = 0, 1, \dots, m$, to estimate the value of m based on a sequential testing procedure. For each k , the corresponding null hypothesis H_0^k is

H_0^k : The number of nonzero components of the coefficient β is k .

which is equivalent to

H_0^k : There exist some indices $1 \leq i_1^* < \dots < i_k^* \leq p$ such that $\beta_{i_1^*} \neq 0, \dots, \beta_{i_k^*} \neq 0$ and $\beta_j = 0$ for any $j \notin \{i_1^*, \dots, i_k^*\}$,

where the superscript k indicates the number of associated portfolios under the null hypothesis H_0^k . In this way, we can see that H_0^0 equals to H_{00} and the analysis of H_0^1 is equivalent to investigating $H_{01}, \dots, H_{0p}$. Motivated by the above test statistics for pleiotropy, we propose a test statistic T_n^k , which is applied to test the null hypothesis H_0^k , i.e.

$$T_n^k = \min_{1 \leq i_1 < \dots < i_k \leq p} \tilde{\theta}_{V_{i_1, \dots, i_k}}^\top A_n \tilde{\theta}_{V_{i_1, \dots, i_k}}, \quad (7)$$

where $\tilde{\theta}_{V_{i_1, \dots, i_k}}$ is the estimator given in Theorem 3.1 under the constraints $V_{i_1, \dots, i_k} \theta_0 = \mathbf{0}$. Moreover, the indices $i_1, \dots, i_k$ are from all possible subsets of choosing k unconstrained β 's among p portfolios. And the constraint matrix $V_{i_1, \dots, i_k}$ is constructed by deleting the row that corresponding to nuisance parameter λ and also deleting the $(i_1, \dots, i_k)$ th rows for β 's, which implies that these parameters are not constrained to zero. For example, when $k = 0$, $T_n^0 = T_{n0}$; when $k = 1$, $T_n^1 = \min_{k=1, \dots, p} T_{nk}$. By using the similar theoretical techniques as Theorem 3.2, we obtain the asymptotic properties of T_n^k as follows.

THEOREM 3.3 Assume that Conditions (C1)–(C4) hold. For fixed $k \in \{0, 1, \dots, p\}$, we can obtain the following conclusions:

- (1) if H_0^k holds, that is, there are k portfolios that are affected by factors, it holds that $T_n^k \xrightarrow{d} \chi_{p-k}^2$ as $n \rightarrow \infty$.
- (2) if H_0^k does not hold, it holds that $T_n^k = O_p(n)$.

Based on the above properties, we propose the following sequential testing procedure. Let α represent the significance level.

Step 1. To test the null hypothesis of no associated portfolios H_0^0 , we compute the statistic T_n^0 and compare the test statistic with critical value $\chi_p^2(\alpha)$. If $T_n^0 \leq \chi_p^2(\alpha)$, the null H_0^0 cannot be rejected, thus we have to accept

the null hypothesis of no associated portfolios. But if $T_n^0 > \chi_p^2(\alpha)$, we would reject H_0^0 and turn to the second step.

Step 2. Consider the null hypothesis H_0^1 , that is, there is only one associated portfolio, and calculate the test statistic $T_n^1 = \min_{k=1, \dots, p} T_{nk}$. If $T_n^1 \leq \chi_{p-1}^2(\alpha)$, we cannot reject H_0^1 , then the test for only one associated portfolio H_0^1 also cannot be rejected. In contrast, if $T_n^1 > \chi_{p-1}^2(\alpha)$, we reject the null hypothesis H_0^1 , then the test for only one associated portfolio would be rejected. In order to identify how many and which components of β are nonzero, we would turn to the next step to test H_0^2 and so on.

Step 3. Similar procedures are continuing until H_0^{m-1} is rejected and H_0^m fails to reject, then stop testing and conclude there are m portfolios associated with the factor. The corresponding test statistic for H_0^m is T_n^m . Thus if $T_n^m \leq \chi_{p-m}^2(\alpha)$, we cannot reject H_0^m . Otherwise, H_0^m would be rejected.

This general sequential procedure systematically introduces a formal way to determine the number of portfolios associated with a factor. To determine which portfolios are associated, the indices $(i_1^*, \dots, i_m^*)$ are determined by the following equation,

$$(i_1^*, \dots, i_m^*) = \arg \min_{1 \leq i_1 < \dots < i_m \leq p} \tilde{\theta}_{V_{i_1, \dots, i_m}}^\top A_n \tilde{\theta}_{V_{i_1, \dots, i_m}}, \quad (8)$$

which illustrates the best model fit. Therefore these indices correspond to the portfolios that are associated.

REMARK 3.2 The *GRS* statistic tests the null hypothesis that all intercepts are zero, which corresponds to H_{00} in our paper. In the appendix, we show that our test statistic T_{n0} and the *GRS* test statistic T_{GRS} are equivalent under the linear framework with the normality assumption. However, the *GRS* test has limitations in detecting cases with non-zero intercepts, a scenario where our test demonstrates a clear advantage. Moreover, our test is not limited to identifying non-zero intercepts; it can also be employed to detect influential factors, making it a more versatile and practical tool for analysis.

4. Numerical studies

In this section, we examine the finite sample performance of our proposed tests in two scenarios. In Section 4.1, a simple case of linear asset pricing model is considered. And we compare the performance of *GRS* with our proposed tests. In Section 4.2, to explore the model performance of our sequential test, we consider a linear model with correlated factors to study the accuracy of identifying the true number of nonzero intercepts. In Section A.7 of the appendix, we also investigate the performance of the 1-order vector autoregression model (VAR(1)) developed by Christopher (1980), which is often used in macroeconomics.

4.1. Scenario 1: comparison with *GRS* test

To investigate the application of our proposed method on the research about asset pricing, we consider two stages: first, we

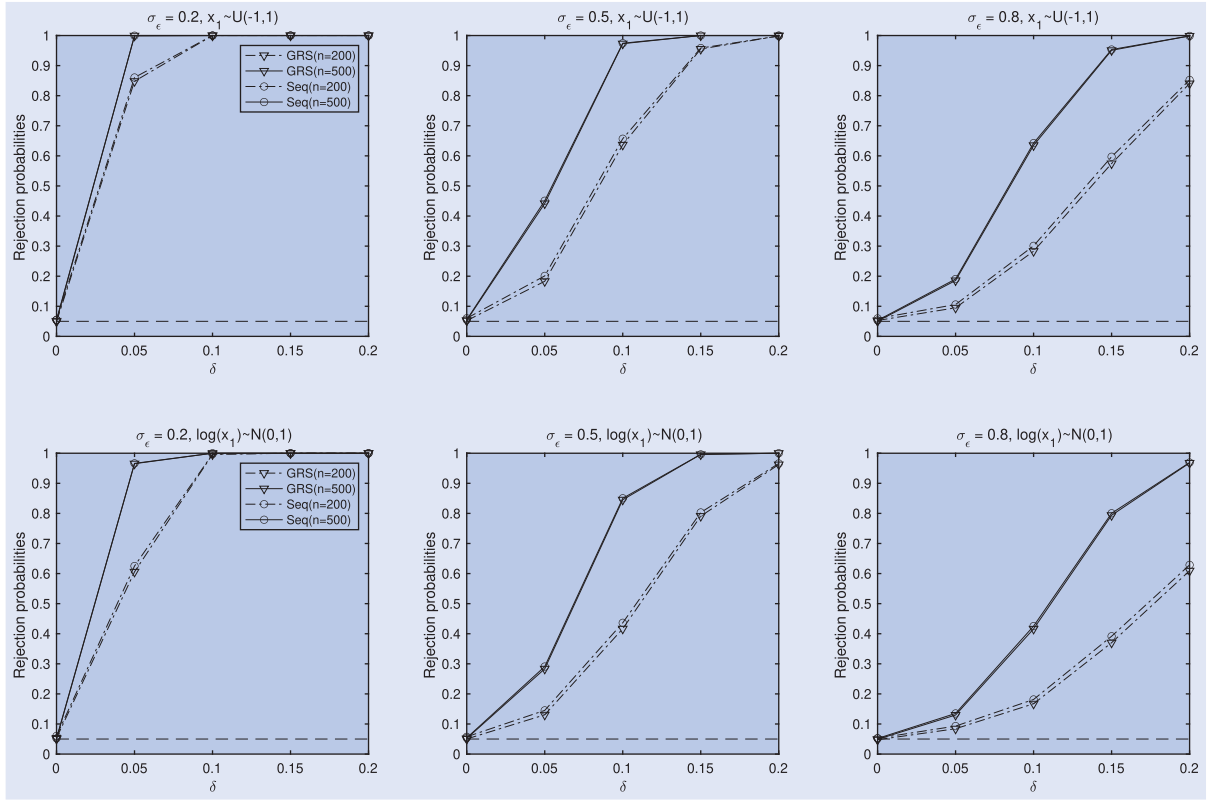


Figure 1. Comparison of GRS test and Sequential test.

are going to demonstrate when testing the null hypothesis that all intercepts of the portfolio-level regressions are zero, our proposed method should at least have the same power as *GRS* test; second, our proposed method could make a further step by providing additional and useful information regarding the efficacy of different factor models for asset returns beyond existing statistics dealing with asset pricing. In this subsection, we focus on the first stage, that is to compare the power of our proposed method with *GRS* test. And the second step could be seen from the simulation study of accuracy rate in Sections 4.2 and A.7 of the appendix.

As stated, our proposed method could also be used to test the null hypothesis that a single factor (the intercept) have no influence on any portfolio (the response). though this is only one special case for our method. Note that this null hypothesis is similar to that of *GRS* test. Thus we choose T_{n0} as the test statistic to compare with *GRS* test, that is $T_{Seq} = T_{n0}$ in this case. Now we conduct a corresponding simulation study for a linear regression model. Assume the model is

$$\begin{cases} y_1 = \delta + x_1 + x_2 + \varepsilon \\ y_2 = x_1 + 2x_2 + \varepsilon \\ y_3 = x_1 + 3x_2 + \varepsilon, \end{cases} \quad (9)$$

where $x_1 \sim U(-b_1, b_1)$ or $\log(x_1) \sim N(0, b_2^2)$, $x_2 \sim N(0, 1)$ and $\varepsilon \sim N(0, \sigma_\varepsilon^2)$. The parameter δ represents the distance between the null and alternative hypothesis, i.e. $\delta = 0$ corresponds to the null, and others correspond to the alternatives. To evaluate the performance on size and power, let $\delta \in \{0, 0.05, 0.1, 0.15, 0.2\}$, $b_1 \in \{1, 3, 5\}$, $b_2 \in \{1, 1.5, 2\}$ and

$\sigma_\varepsilon \in \{0.2, 0.5, 0.8\}$. The sample size is $T = 200, 500$ and each trial repeats 10 000 times.

We now report the results of the empirical sizes and power by T_{GRS} and T_{Seq} in figure 1 and tables 1–2. From figure 1, the following conclusions can be drawn. When $\delta = 0$, our proposed method and *GRS* test are both close to the significance level 0.05. And with the growth of sample size and distance away from the null hypothesis, the power of both two methods increase. This verifies the consistency of these methods. Besides, our proposed method T_{Seq} has slightly higher power than T_{GRS} . Regarding the value of standard deviation of residuals σ_ε , we observe a nearly monotonic decrease of the rejection rates with increasing σ_ε . This is not surprising, since larger σ_ε makes more uncertainty, which leads to worse efficiency of the test. For more cases' comparison, we present the results in tables 1–2. Similar conclusions are obtained.

4.2. Scenario 2: sequential testing

Consider the model as follows,

$$\mathbf{y}_t = \mathbf{c} + \beta_1 x_{t1} + \beta_2 x_{t2} + \beta_3 x_{t3} + \boldsymbol{\varepsilon}_t, \quad t = 1, \dots, T, \quad (10)$$

where $\mathbf{y}_t$ is a p -dimensional random vector, x_{t1}, x_{t2}, x_{t3} are scalar variables and $\boldsymbol{\varepsilon}_t \stackrel{i.i.d}{\sim} N(\mathbf{0}, \Sigma^\varepsilon)$ with $\Sigma^\varepsilon = (\sigma_{jk}^\varepsilon)_{p \times p}$. $\mathbf{c} = (c_1, \dots, c_p)^\top$ denotes the intercepts, and $\mathbf{c}, \beta_1, \beta_2$ and β_3 are p -dimensional vectors. Then $\mathbf{c}, \beta_1, \beta_2, \beta_3$ and Σ^ε are the parameters of the model. Let $p \in \{6, 10\}$. We conduct this simulation to evaluate the accuracy of choosing the correct number of nonzero c_j 's. For example, if $p = 6$, let the true

Table 1. Rejection probabilities for the GRS test T_{GRS} and the sequential test T_{Seq} (T_{n0}) at 5% significance level when $x_1 \sim U(-b_1, b_1)$.

T	δ	$b_1 = 1$		$b_1 = 3$		$b_1 = 5$	
		T_{GRS}	T_{Seq}	T_{GRS}	T_{Seq}	T_{GRS}	T_{Seq}
200				$\sigma_\varepsilon = 0.2$			
	0	0.0491	0.0548	0.0533	0.0591	0.0480	0.0550
	0.05	0.8481	0.8605	0.8430	0.8557	0.8470	0.8595
	0.10	1	1	1	1	1	1
	0.15	1	1	1	1	1	1
	0.20	1	1	1	1	1	1
				$\sigma_\varepsilon = 0.5$			
	0	0.0522	0.0601	0.0464	0.0522	0.0481	0.0567
	0.05	0.1831	0.2010	0.1900	0.2064	0.1908	0.2057
	0.10	0.6367	0.6569	0.6353	0.6549	0.6405	0.6587
	0.15	0.9552	0.9583	0.9539	0.9582	0.9581	0.9624
	0.20	0.9985	0.9986	0.9989	0.9991	0.9987	0.9994
				$\sigma_\varepsilon = 0.8$			
	0	0.0529	0.0597	0.0484	0.0545	0.0522	0.0584
	0.05	0.0949	0.1059	0.0977	0.1098	0.0968	0.1065
	0.10	0.2831	0.3009	0.2803	0.2998	0.2914	0.3111
	0.15	0.5745	0.5970	0.5863	0.6031	0.5814	0.6022
	0.20	0.8404	0.8523	0.8390	0.8530	0.8521	0.8653
500				$\sigma_\varepsilon = 0.2$			
	0	0.0514	0.0546	0.0487	0.0512	0.0518	0.0543
	0.05	0.9992	0.9994	0.9991	0.9995	0.9982	0.9984
	0.10	1	1	1	1	1	1
	0.15	1	1	1	1	1	1
	0.20	1	1	1	1	1	1
				$\sigma_\varepsilon = 0.5$			
	0	0.0531	0.0553	0.0519	0.0549	0.0495	0.0518
	0.05	0.4418	0.4508	0.4422	0.4505	0.4386	0.4490
	0.10	0.9729	0.9744	0.9710	0.9721	0.9732	0.9750
	0.15	1	1	1	1	1	1
	0.20	1	1	1	1	1	1
				$\sigma_\varepsilon = 0.8$			
	0	0.0505	0.0531	0.0550	0.0576	0.0463	0.0484
	0.05	0.1857	0.1906	0.1858	0.1928	0.1848	0.1906
	0.10	0.6349	0.6424	0.6399	0.6479	0.6397	0.6485
	0.15	0.9508	0.9534	0.9545	0.9570	0.9554	0.9572
	0.20	0.9986	0.9987	0.9993	0.9993	0.9991	0.9992

number of nonzero c_j 's varies from 1 to 6, then we record the estimated numbers from 10 000 simulations of our proposed method, and compute the frequency. Let the nonzero intercept equals to 1, the marginal effect of the coefficients $\beta_1 = \mathbf{1}_p$, $\beta_2 = 0.5\mathbf{1}_p$ and $\beta_3 = 0.3\mathbf{1}_p$, where $\mathbf{1}_p$ denotes a p -dimensional vector with components equaling to 1. Assume that $(x_{i1}, x_{i2}, x_{i3})^\top \stackrel{i.i.d.}{\sim} N(\mathbf{0}, \Sigma^x)$, where $\Sigma^x = (\sigma_{jk}^x)_{3 \times 3}$ with $\sigma_{jj}^x = 1$ and $\sigma_{jk}^x = 0.5$ for $j \neq k$. Let $\sigma_{jj}^\varepsilon = 1$, $\sigma_{jk}^\varepsilon = 0.2$ for $j \neq k$, the sample size $T = 500$ and the significance level $\alpha \in \{0.01, 0.05\}$.

The results in tables 3–4 show that the accuracy rates are near to $1 - \alpha$, which implies that the sequential testing procedure is powerful. The false discovery probabilities are controlled by the significance level. And the false estimated numbers of nonzero c_j 's are still close to the true numbers. In general case, sequential testing procedure could select the number of nonzero intercepts relatively effectively and correctly.

5. Data analysis

In this section, we conduct real data analysis to investigate whether our proposed method could indeed provide additional and useful information regarding the efficacy of different factor models, compared to *GRS*. Our baseline is Fama-French five factor model, in which, returns can be accounted for directly by factors, i.e. market, size, value, profitability, and investment pattern, through a linear function, details as follows,

$$R_{it} - R_{Ft} = \alpha_i + b_i(R_{Mt} - R_{Ft}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + \varepsilon_{it}. \quad (11)$$

Here, R_{it} is the return on security or portfolio i for period t , R_{Ft} is the risk-free return, R_{Mt} is the return on the value-weight (VW) market portfolio, and thus $R_{it} - R_{Ft}$ and $R_{Mt} - R_{Ft}$ measure the excess returns on the corresponding security or

Table 2. Rejection probabilities for the GRS test T_{GRS} and the sequential test T_{Seq} (T_{n0}) at 5% significance level when $\log(x_1) \sim N(0, b_2)$.

T	δ	$b_2 = 1$		$b_2 = 1.5$		$b_2 = 2$	
		T_{GRS}	T_{Seq}	T_{GRS}	T_{Seq}	T_{GRS}	T_{Seq}
200				$\sigma_\varepsilon = 0.2$			
	0	0.0534	0.0603	0.0486	0.0542	0.0515	0.0583
	0.05	0.6057	0.6251	0.7610	0.7751	0.8033	0.8186
	0.10	0.9962	0.9971	0.9998	0.9999	0.9999	0.9999
	0.15	1	1	1	1	1	1
	0.20	1	1	1	1	1	1
				$\sigma_\varepsilon = 0.5$			
	0	0.0502	0.0574	0.0491	0.0547	0.0498	0.0568
	0.05	0.1305	0.1452	0.1580	0.1719	0.1691	0.1845
	0.10	0.4166	0.4363	0.5509	0.5724	0.6046	0.6262
	0.15	0.7910	0.8031	0.9032	0.9110	0.9276	0.9359
	0.20	0.9623	0.9660	0.9939	0.9948	0.9978	0.9980
				$\sigma_\varepsilon = 0.8$			
	0	0.0477	0.0527	0.0483	0.0555	0.0502	0.0577
	0.05	0.0842	0.0930	0.0906	0.0993	0.0902	0.0998
	0.10	0.1672	0.1815	0.2261	0.2443	0.2552	0.2726
	0.15	0.3702	0.3916	0.4773	0.5001	0.5352	0.5590
	0.20	0.6093	0.6287	0.7580	0.7727	0.8067	0.8214
500				$\sigma_\varepsilon = 0.2$			
	0	0.0502	0.0522	0.0505	0.0526	0.0496	0.0520
	0.05	0.9652	0.9663	0.9956	0.9958	0.9989	0.9990
	0.10	1	1	1	1	1	1
	0.15	1	1	1	1	1	1
	0.20	1	1	1	1	1	1
				$\sigma_\varepsilon = 0.5$			
	0	0.0519	0.0539	0.0548	0.0575	0.0509	0.0548
	0.05	0.2834	0.2909	0.3751	0.3827	0.4214	0.4294
	0.10	0.8453	0.8502	0.9425	0.9454	0.9654	0.9670
	0.15	0.9960	0.9962	1	1	0.9999	0.9999
	0.20	1	1	1	1	1	1
				$\sigma_\varepsilon = 0.8$			
	0	0.0491	0.0519	0.0511	0.0534	0.0490	0.0516
	0.05	0.1298	0.1351	0.1608	0.1666	0.1743	0.1808
	0.10	0.4161	0.4254	0.5609	0.5686	0.6145	0.6213
	0.15	0.7938	0.8004	0.9120	0.9154	0.9357	0.9374
	0.20	0.9677	0.9691	0.9943	0.9949	0.9980	0.9981

portfolio and market portfolio. SMB_t (*Size* factor) is the return on a diversified portfolio of small stocks minus the return on a diversified portfolio of big stocks, HML_t is the difference between the returns on diversified portfolios of high and low book-to-market stocks, RMW_t is the difference between the returns on diversified portfolios of stocks with robust and weak profitability, CMA_t is the difference between the returns on diversified portfolios of the stocks of low and high investment firms, which we call conservative and aggressive, and ε_{it} is a zero-mean error term. In details, we apply our proposed method to the data set of average monthly returns for portfolios and five factors, lasting from July 1963 to July 2024, which extends the data sample of Fama and French (2015) for 11 more years. The data set is constructed by Fama and French and is freely downloadable from the website of

Kenneth French Data Library[†]. The data set covers all nonfinancial firms in the intersection of (a) the NYSE, AMEX, and NASDAQ return files from the Center for Research in Security Prices (CRSP) and (b) the merged COMPUSTAT annual industrial files of income-statement and balance-sheet data, also maintained by CRSP.

For factors construction, we use the 2×3 sorts on *Size* and Book-to-Market (B/M), *Size* and operating profitability (OP), or *Size* and investment (Inv). Independent sorts of stocks are assigned to two *Size* groups (based on NYSE median), and three B/M (based on 30th and 70th NYSE percentiles), OP (based on 30th and 70th NYSE percentiles), and Inv (based

[†] http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Table 3. Sequential testing for the accuracy of identifying the true number of nonzero c_j 's when $T = 500$ and $p = 6$.

True No.	Est.1	Est.2	Est.3	Est.4	Est.5	Est.6
Estimated number of nonzero c_j 's (significance level: 0.05)						
True.1	0.9432	0.0527	0.0034	0.0006	0.0001	0
True.2	0	0.9479	0.0474	0.0043	0.0004	0
True.3	0	0	0.9494	0.0468	0.0029	0.0009
True.4	0	0	0	0.9503	0.0462	0.0035
True.5	0	0	0	0	0.9465	0.0535
True.6	0	0	0	0	0	1
Estimated number of nonzero c_j 's (significance level 0.01)						
True.1	0.9889	0.0109	0.0001	0.0001	0	0
True.2	0	0.9892	0.0106	0.0002	0	0
True.3	0	0	0.9896	0.0102	0.0002	0
True.4	0	0	0	0.9897	0.0102	0.0001
True.5	0	0	0	0	0.9886	0.0114
True.6	0	0	0	0	0	1

Table 4. Sequential testing for the accuracy of identifying the true number of nonzero c_j 's when $T = 500$ and $p = 10$.

True No.	Est.1	Est.2	Est.3	Est.4	Est.5	Est.6	Est.7	Est.8	Est.9	Est.10
Estimated number of nonzero c_j 's (significance level: 0.05)										
True.1	0.9413	0.0536	0.0046	0.0003	0.0002	0	0	0	0	0
True.2	0	0.9408	0.0542	0.0047	0.0003	0	0	0	0	0
True.3	0	0	0.9423	0.0521	0.0046	0.0008	0.0001	0.0001	0	0
True.4	0	0	0	0.9510	0.0451	0.0036	0.0003	0	0	0
True.5	0	0	0	0	0.9431	0.0517	0.0044	0.0008	0	0
True.6	0	0	0	0	0	0.9492	0.0468	0.0035	0.0004	0.0001
True.7	0	0	0	0	0	0	0.9468	0.0497	0.0026	0.0009
True.8	0	0	0	0	0	0	0	0.9485	0.0481	0.0034
True.9	0	0	0	0	0	0	0	0	0.9565	0.0435
True.10	0	0	0	0	0	0	0	0	0	1
Estimated number of nonzero c_j 's (significance level: 0.01)										
True.1	0.9878	0.0117	0.0005	0	0	0	0	0	0	0
True.2	0	0.9870	0.0126	0.0004	0	0	0	0	0	0
True.3	0	0	0.9885	0.0107	0.0008	0	0	0	0	0
True.4	0	0	0	0.9885	0.0112	0.0003	0	0	0	0
True.5	0	0	0	0	0.9890	0.0107	0.0003	0	0	0
True.6	0	0	0	0	0	0.9897	0.0100	0.0003	0	0
True.7	0	0	0	0	0	0	0.9889	0.0109	0.0002	0
True.8	0	0	0	0	0	0	0	0.9894	0.0105	0.0001
True.9	0	0	0	0	0	0	0	0	0.9891	0.0109
True.10	0	0	0	0	0	0	0	0	0	1

on 30th and 70th NYSE percentiles) groups. *SMB* factor is thus defined by the average *SMB* factors constructed using sorts on *Size* and *B/M*, *Size* and *OP*, and *Size* and *Inv*. For instance, for sorts on *Size* and *B/M*, *SMB* is defined by the different average returns between portfolio with small stocks and big stocks, and average here means value-weighted average returns of stocks based on market caps with high, neutral, and low *B/M* within portfolio of small/big stocks. For *HML*, *RMW*, and *CMA* factors, they are constructed using the different average returns between portfolio with high *B/M* (robust profitability, or conservative investment) stocks and low *B/M* (weak profitability, or aggressive investment) stocks, and average here means arithmetic average returns of small and big stocks within portfolios (details in table 5). To support the i.i.d. normality assumption, we have included boxplots and QQ-plots of the portfolio returns. Specifically, figure 2 presents the boxplots, which visually depict the distribution of the data. The

boxplots indicate that the portfolio returns are nearly symmetrical. Due to space limitations, we provide the QQ-plots for the residuals of 25 *Size-B/M* portfolios as examples in figure 3, while the remaining QQ-plots are included in the Appendix. These plots collectively suggest that the normality assumption is reasonable.

In table 6, similar to Fama and French (2015), we estimate (11) and consider five asset pricing models: (i) the three-factor models that combine excess return $R_{Mt} - R_{Ft}$ and *SMB* with *HML*; (ii) the four-factor models that combine $R_{Mt} - R_{Ft}$, *SMB*, and pairs of *HML*, *RMW*, and *CMA*; and (iii) the five-factor model. We test how well the three-, four-, and five-factor models explain average monthly excess returns on six sets of left-hand-side portfolios, that is, 25 *Size-B/M* portfolios (Panel A), 25 *Size-OP* portfolios (Panel B), 25 *Size-Inv* portfolios (Panel C), 32 *Size-B/M-OP* portfolios (Panel D), 32 *Size-B/M-Inv* portfolios (Panel E), and 32

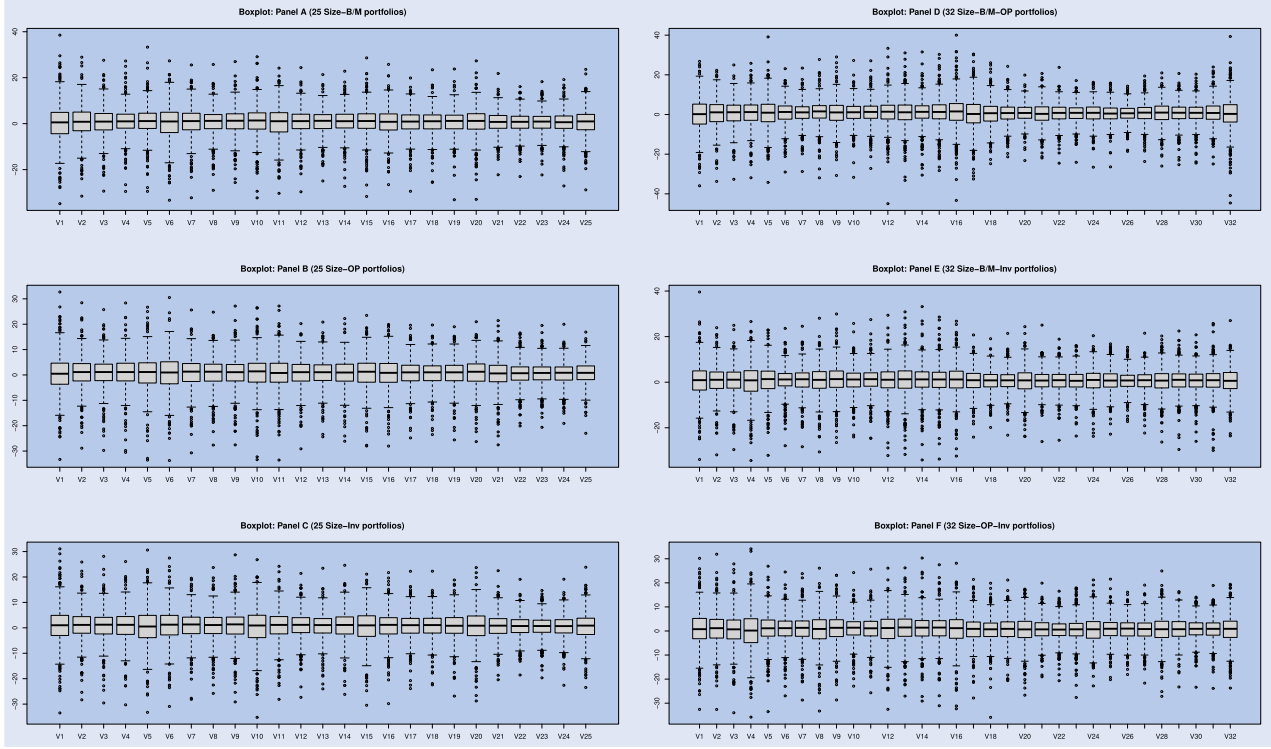


Figure 2. Boxplots of the six types of portfolios from July 1963–July 2024.

Table 5. Construction of *Size*, *B/M*, profitability (*OP*), and investment (*Inv*) factors.

Sort	Breakpoints	Factors and their components
2 × 3 sorts on <i>Size</i> and <i>B/M</i> , or <i>Size</i> and <i>OP</i> , or <i>Size</i> and <i>Inv</i>	<i>Size</i> : NYSE median B/M: 30th and 70th NYSE percentiles OP: 30th and 70th NYSE percentiles Inv: 30th and 70th NYSE percentiles	$SMB = (SMB_{B/M} + SMB_{OP} + SMB_{Inv})/3$ $SMB_{B/M} = (SH + SN + SL)/3 - (BH + BN + BL)/3$ $SMB_{OP} = (SR + SN + SW)/3 - (BR + BN + BW)/3$ $SMB_{Inv} = (SC + SN + SA)/3 - (BC + BN + BA)/3$ $HML = (SH + BH)/2 - (SL + BL)/2 = [(SH - SL) + (BH - BL)]/2$ $RMW = (SR + BR)/2 - (SW + BW)/2 = [(SR - SW) + (BR - BW)]/2$ $CMA = (SC + BC)/2 - (SA + BA)/2 = [(SC - SA) + (BC - BA)]/2$

Notes: We use independent sorts to assign stocks to two *Size* groups, and three *B/M*, *OP*, and *Inv* groups. The value-weighted portfolios defined by the intersections of the groups are the building blocks for the factors. We label these portfolios with two letters. The first always describes the *Size* group, small (*S*) or big (*B*), and the second describes the *B/M* group, high (*H*), neutral (*N*), or low (*L*), the *OP* group, robust (*R*), neutral (*N*), or weak (*W*), or the *Inv* group, conservative (*C*), neutral (*N*), or aggressive (*A*). The factors are *SMB* (small minus big), *HML* (high minus low *B/M*), *RMW* (robust minus weak *OP*), and *CMA* (conservative minus aggressive *Inv*).

Size-OP-Inv portfolios (Panel F), based on several statistics. Certainly, there are a few other ad hoc factor models but they are less commonly adopted in practice.

To be specific, if an asset pricing model completely captures expected returns, the intercept should be indistinguishable from zero in a regression of an asset's excess returns on the model's factor returns. In table 6, we report five statistics for the five asset pricing models with six sets of portfolios mentioned above. Among them, four statistics are reported by Fama and French (2015): (1) *GRS* statistic with *p*-value, which directly tests the hypothesis that all intercepts of regressions in portfolio-level are indistinguishable from zero; (2) $A|a_i|$, which is the average absolute value of all intercepts, measuring the dispersion of the intercepts produced by a given model for a set of LHS portfolios and should be close to zero if the model largely captures expected returns; (3) $A|a_i|/A|\bar{r}_i|$, which is the ratio of the average absolute value of the intercept a_i to the average absolute value of $\bar{r}_i$, the

average return on portfolio i ($\bar{r}_i$) minus the average of the portfolio returns ($\bar{R}$). This ratio estimates the proportion of the cross-sectional variation in expected returns left unexplained by the model, with lower values indicating a better fit; (4) $A(\hat{\sigma}_i^2)/A(\hat{u}_i^2)$, which is the ratio of the average squared intercept to the average squared value of $\hat{u}_i$, corrected for sampling error in the numerator and denominator. This statistic also measures the proportion of explained variation in expected returns, with lower values indicating better explanatory power. Finally, the fifth statistic, k , calculates the number of intercepts that are statistically (at a 5 percent level of significance) non-zero based on the pleiotropy test proposed in this paper. Obviously, the asset pricing model explains average monthly excess returns well if k is close to zero. In table 7, we report the corresponding *p*-values for the tests of three-, four-, and five-factor models.

Relative to traditional test statistics for asset pricing models, the greatest advantage of the pleiotropy test lies in its

Table 6. Summary statistics for tests of three-, four-, and five-factor models; July 1963-July 2024, 733 months.

GRS (<i>p</i> -value)	<i>A</i> <i>a_i</i>	$\frac{A a_i }{A \bar{r}_i }$	$\frac{A(\hat{a}_i^2)}{A(\hat{u}_i^2)}$	<i>k</i>	GRS (<i>p</i> -value)	<i>A</i> <i>a_i</i>	$\frac{A a_i }{A \bar{r}_i }$	$\frac{A(\hat{a}_i^2)}{A(\hat{u}_i^2)}$	<i>k</i>		
Panel A: 25 Size-B/M portfolios					Panel D: 32 Size-B/M-OP portfolios						
HML	4.0317 (3.26E-10)	0.1027	0.6894	0.6451	6	HML	2.5211 (1.04E-05)	0.1367	0.6700	0.5016	5
HML,RMW	3.4929 (3.03E-08)	0.0959	0.6439	0.4655	5	HML,RMW	1.8557 (0.0031)	0.1032	0.5062	0.2958	2
HML,CMA	3.8820 (1.16E-09)	0.1011	0.6787	0.6072	6	HML,CMA	2.7450 (1.28E-06)	0.1473	0.7220	0.5329	6
RMW,CMA	3.3099 (1.37E-07)	0.1021	0.6851	0.5006	4	RMW,CMA	1.8679 (0.0028)	0.1189	0.5829	0.3308	2
HML,RMW,CMA	3.2824 (1.72E-07)	0.0923	0.6195	0.4315	4	HML,RMW,CMA	1.8756 (0.0027)	0.1120	0.5493	0.2894	3
Panel B: 25 Size-OP portfolios					Panel E: 32 Size-B/M-Inv portfolios						
HML	2.6998 (1.75E-05)	0.1048	0.7650	0.6982	4	HML	2.9529 (1.72E-07)	0.1231	0.7690	0.5786	8
HML,RMW	2.0549 (0.0019)	0.0684	0.4995	0.2710	2	HML,RMW	2.5584 (7.39E-06)	0.1196	0.7473	0.5739	8
HML,CMA	3.0093 (1.55E-06)	0.1179	0.8609	0.8317	4	HML,CMA	2.9193 (2.39E-07)	0.1012	0.6321	0.4740	5
RMW,CMA	2.0532 (0.0019)	0.0735	0.5369	0.3137	2	RMW,CMA	2.2673 (0.0001)	0.1072	0.6698	0.5272	3
HML,RMW,CMA	2.0911 (0.0015)	0.0678	0.4950	0.2691	2	HML,RMW,CMA	2.4100 (2.88E-05)	0.0979	0.6117	0.4339	3
Panel C: 25 Size-Inv portfolios					Panel F: 32 Size-OP-Inv portfolios						
HML	4.5232 (4.82E-12)	0.0992	0.7224	0.7155	6	HML	3.7712 (4.47E-11)	0.1640	0.8907	0.7772	13
HML,RMW	3.9990 (4.32E-10)	0.0897	0.6535	0.6480	5	HML,RMW	3.0507 (6.59E-08)	0.1110	0.6026	0.3932	7
HML,CMA	4.1364 (1.34E-10)	0.0961	0.6997	0.6132	6	HML,CMA	3.3459 (3.47E-09)	0.1556	0.8453	0.7272	10
RMW,CMA	3.3303 (1.16E-07)	0.0830	0.6047	0.4365	4	RMW,CMA	2.4256 (2.50E-05)	0.0939	0.5102	0.2624	4
HML,RMW,CMA	3.3435 (1.04E-07)	0.0841	0.6124	0.4387	5	HML,RMW,CMA	2.4395 (2.20E-05)	0.0931	0.5054	0.2591	5

Notes: The table tests the ability of three-, four-, and five-factor models to explain monthly excess returns on 25 *Size-B/M portfolios* (Panel A), 25 *Size-OP portfolios* (Panel B), 25 *Size-Inv portfolios* (Panel C), 32 *Size-B/M-OP portfolios* (Panel D), 32 *Size-B/M-Inv portfolios* (Panel E), and 32 *Size-OP-Inv portfolios* (Panel F). For each set of 25 or 32 regressions, the table shows the factors that augment $R_M - R_F$ and SMB in the regression model, the *GRS* statistic testing whether the expected values of all 25 or 32 intercept estimates are zero, the average absolute value of the intercepts, $A|a_i|$, $A|a_i|/A|\bar{r}_i|$, the average absolute value of the intercept a_i over the average absolute value of $\bar{r}_i$, which is the average return on portfolio i minus the average of the portfolio returns, and $A(\hat{a}_i^2)/A(\hat{u}_i^2)$, which is $A(a_i^2)/A(\bar{r}_i^2)$, the average squared intercept over the average square value of $\bar{r}_i$, corrected for sampling error in the numerator and denominator, and finally and most importantly, k indicates how many intercepts are statistically non-zero by our proposed method.

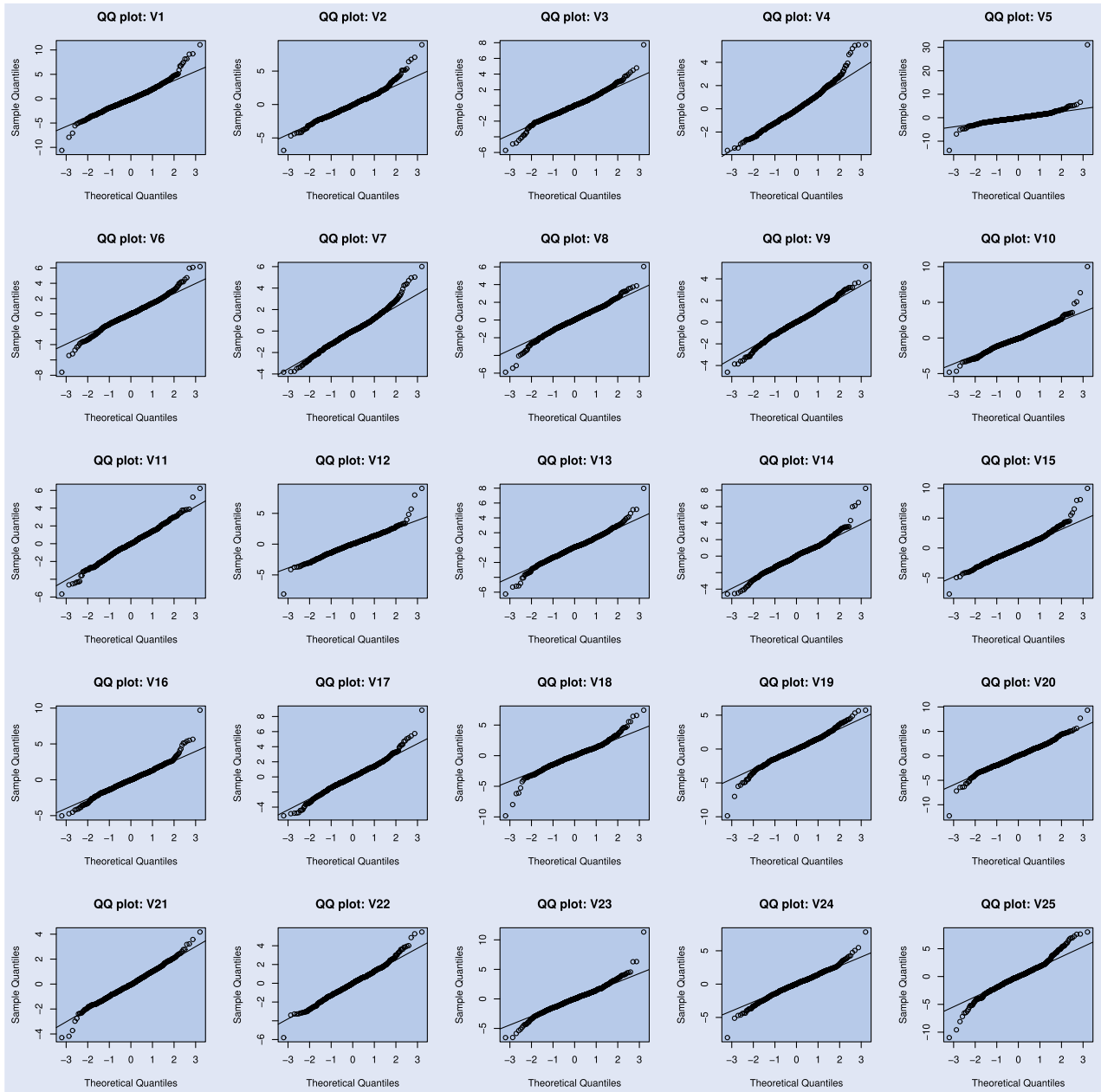


Figure 3. QQ-plots for 25 Size-B/M portfolios.

ability to provide a direct count of intercepts that deviate significantly from zero, thereby offering a more intuitive measure of model inadequacy. For instance, in Row (1) of Panel A in table 6, when using the Fama-French three-factor model to explain monthly excess returns on 25 Size-B/M portfolios, we observe that 6 out of 25 portfolios have non-zero intercepts, indicating that the model fails to fully account for the expected returns on these portfolios and may require additional factors.

While $A|a_i|$, $A|a_i|/A|\bar{r}_i|$, and $A(\hat{\alpha}_i^2)/A(\hat{u}_i^2)$ provide useful summary statistics on the overall goodness-of-fit, they suffer from significant limitations compared to the pleiotropy test based on k . First, these statistics aggregate information across all portfolios and thus may obscure cases where only a subset of portfolios are poorly explained. For example, in Panel D of table 6, we see that $A|a_i|/A|\bar{r}_i|$ and $A(\hat{\alpha}_i^2)/A(\hat{u}_i^2)$ are relatively low for the model with factors *SMB*, *HML*, and *RMW*, suggesting an overall good fit. However, the k statistic reveals

that 2 out of the 32 portfolios still have statistically significant non-zero intercepts, indicating that the model does not adequately explain the returns of these particular portfolios despite the favorable values of the average-based statistics.

Moreover, the average-based statistics lack clear threshold values for determining model adequacy. While lower values of $A|a_i|$, $A|a_i|/A|\bar{r}_i|$, and $A(\hat{\alpha}_i^2)/A(\hat{u}_i^2)$ may suggest a better fit, there is no consensus on what constitutes an acceptable value, making it difficult to draw definitive conclusions about model performance. In contrast, the k statistic directly counts the number of portfolios for which the intercept is significantly different from zero, providing a straightforward benchmark for assessing model fit. This directness allows for easier interpretation, as demonstrated in Panel F of table 6, where the five-factor model applied to 32 Size-OP-Inv portfolios produces a k value of 3, suggesting that most of the portfolios are well-explained by the model, even if the *GRS*

Table 7. Corresponding p -values for tests of three-, four-, and five-factor models.

	$k - 1$	$T_n^{k-1}(p\text{-value})$	k	$T_n^k(p\text{-value})$		$k - 1$	$T_n^{k-1}(p\text{-value})$	k	$T_n^k(p\text{-value})$
<i>Panel A: 25 Size-B/M portfolios</i>					<i>Panel D: 32 Size-B/M-OP portfolios</i>				
HML	5	37.5235 (0.0101)	6	29.5718 (0.0575)	HML	4	41.6383 (0.0469)	5	35.9574 (0.1162)
HML,RMW	4	34.2833 (0.0338)	5	29.5292 (0.0779)	HML,RMW	1	46.4719 (0.0367)	2	40.3383 (0.0985)
HML,CMA	5	35.4566 (0.0178)	6	28.0658 (0.0822)	HML,CMA	5	42.6892 (0.0281)	6	38.5780 (0.0535)
RMW,CMA	3	39.7303 (0.0116)	4	32.2317 (0.0555)	RMW,CMA	1	47.8105 (0.0274)	2	41.0071 (0.0868)
HML,RMW,CMA	3	39.3510 (0.0129)	4	31.9444 (0.0593)	HML,RMW,CMA	2	45.0222 (0.0384)	3	39.0405 (0.1009)
<i>Panel B: 25 Size-OP portfolios</i>					<i>Panel E: 32 Size-B/M-Inv portfolios</i>				
HML	3	34.0078 (0.0490)	4	24.5258 (0.2683)	HML	7	40.7883 (0.0242)	8	35.3289 (0.0637)
HML,RMW	1	37.6202 (0.0379)	2	29.1946 (0.1739)	HML,RMW	7	38.6743 (0.0397)	8	34.5976 (0.0746)
HML,CMA	3	38.6931 (0.0153)	4	27.1131 (0.1671)	HML,CMA	4	46.8705 (0.0141)	5	39.7469 (0.0541)
RMW,CMA	1	40.2165 (0.0203)	2	30.6647 (0.1313)	RMW,CMA	2	49.4319 (0.0142)	3	40.2310 (0.0802)
HML,RMW,CMA	1	40.6740 (0.0181)	2	30.4143 (0.1379)	HML,RMW,CMA	2	52.7333 (0.0063)	3	41.9183 (0.0571)
<i>Panel C: 25 Size-Inv portfolios</i>					<i>Panel F: 32 Size-OP-Inv portfolios</i>				
HML	5	33.2415 (0.0317)	6	29.1299 (0.0640)	HML	12	33.0468 (0.0333)	13	29.8352 (0.0539)
HML,RMW	4	35.2733 (0.0263)	5	30.6855 (0.0595)	HML,RMW	6	40.6644 (0.0335)	7	35.8789 (0.0735)
HML,CMA	5	32.8401 (0.0351)	6	28.7997 (0.0692)	HML,CMA	9	37.5681 (0.0283)	10	33.7302 (0.0523)
RMW,CMA	3	35.2297 (0.0367)	4	30.6988 (0.0788)	RMW,CMA	3	47.6937 (0.0158)	4	41.1092 (0.0525)
HML,RMW,CMA	4	33.5008 (0.0410)	5	29.7917 (0.0733)	HML,RMW,CMA	4	41.6336 (0.0469)	5	35.9757 (0.1158)

Notes: The table reports the test results of null hypothesis that there is at most $k - 1$ (or k) intercepts are statistically non-zero for 25 *Size-B/M portfolios* (Panel A), 25 *Size-OP portfolios* (Panel B), 25 *Size-Inv portfolios* (Panel C), 32 *Size-B/M-OP portfolios* (Panel D), 32 *Size-B/M-Inv portfolios* (Panel E), and 32 *Size-OP-Inv portfolios* (Panel F).

Table 8. Corresponding indicies for tests of three-, four-, and five-factor models.

	k	corresponding indices		k	corresponding indices
<i>Panel A: 25 Size-B/M portfolios</i>			<i>Panel D: 32 Size-B/M-OP portfolios</i>		
HML	6	1,4,5,16,21,24	HML	5	1,8,11,20,29
HML,RMW	5	1,5,12,16,24	HML,RMW	2	1,30
HML,CMA	6	1,4,5,16,21,24	HML,CMA	6	1,8,11,13,20,29
RMW,CMA	4	1,12,16,24	RMW,CMA	2	1,11
HML,RMW,CMA	4	1,12,16,24	HML,RMW,CMA	3	1,4,29
<i>Panel B: 25 Size-OP portfolios</i>			<i>Panel E: 32 Size-B/M-Inv portfolios</i>		
HML	4	1,5,22,25	HML	8	3,4,11,20,28,30,31,32
HML,RMW	2	1,5	HML,RMW	8	3,4,9,11,20,28,29,30
HML,CMA	4	1,5,22,25	HML,CMA	5	3,4,11,20,30
RMW,CMA	2	1,5	RMW,CMA	3	3,11,20
HML,RMW,CMA	2	1,5	HML,RMW,CMA	3	3,11,20
<i>Panel C: 25 Size-Inv portfolios</i>			<i>Panel F: 32 Size-OP-Inv portfolios</i>		
HML	6	2,5,8,10,12,15	HML	13	2,3,4,7,8,9,13,16,18,24,27,29,32
HML,RMW	5	1,2,5,10,12	HML,RMW	7	2,4,7,8,16,18,24
HML,CMA	6	5,8,9,10,12,25	HML,CMA	10	3,4,7,8,9,13,15,18,22,32
RMW,CMA	4	1,5,23,25	RMW,CMA	4	4,7,16,18
HML,RMW,CMA	5	1,5,10,23,25	HML,RMW,CMA	5	4,7,16,18,32

Notes: The table reports the portfolios that are not perfectly explained by factor models for 25 *Size-B/M portfolios* (Panel A), 25 *Size-OP portfolios* (Panel B), 25 *Size-Inv portfolios* (Panel C), 32 *Size-B/M-OP portfolios* (Panel D), 32 *Size-B/M-Inv portfolios* (Panel E), and 32 *Size-OP-Inv portfolios* (Panel F). Index i in Panel A, B, and C with 25 *X-Y portfolios* (independent sorts on X and Y factors) refers to the portfolio constructed by stocks from the $[(i - 1)/5] + 1$ quintile of X factor and the $i - 5[(i - 1)/5]$ quintile of Y factor. Index i in Panel D, E, and F with 32 *X-Y-Z portfolios* (independent sorts on X, Y, and Z factors) refers to the portfolio in the $[(i - 1)/16] + 1$ half of X factor, the $[(j - 1)/4] + 1$ quartile of Y factor, and the $j - 4[(j - 1)/4]$ quartile of Z factor, where $j = i - 16[(i - 1)/16]$.

test rejects the null hypothesis at conventional significance levels.

Furthermore, the pleiotropy test offers greater robustness in cases where the efficiency of the *GRS* test is compromised by large residual variances (σ_ϵ), as shown in Rows (2) and (3) of Panel E. Here, the *GRS* test suggests that the model with factors *SMB*, *HML*, and *CMA* is a worse fit than the model with factors *SMB*, *HML*, and *RMW* (with *GRS* values of 2.919 vs. 2.558), yet the pleiotropy test indicates that the former model has fewer portfolios with non-zero intercepts (6 vs. 7). This discrepancy can be attributed to larger σ_ϵ in the

former model, which reduces the power of the *GRS* test, while the pleiotropy test remains effective in identifying the specific portfolios where the model fails.

In conclusion, while traditional statistics such as *GRS*, $A|a_i|$, $A|a_i|/A|\bar{r}_i|$, and $A(\hat{\alpha}_i^2)/A(\hat{u}_i^2)$ provide useful insights, they are limited in their ability to offer precise and interpretable assessments of model performance. The k statistic, derived from our proposed pleiotropy test, addresses these limitations by directly indicating the number of portfolios that remain inadequately explained by the model, thereby offering a more nuanced and robust evaluation of asset pricing models.

Now we could analyze the performance of asset pricing model based on our proposed pleiotropy test. We find that the value of $k(s)$ in all cases are non-zero, suggesting that the widely used factor models are still imperfect interpretation of asset returns. Similar to Fama and French (2015), we are interested in the improvements in descriptions of average returns provided by adding profitability and investment factors to the Fama-French three-factor model. First, for all six sets of LHS portfolios, the five-factor model produces a lower k than the original three-factor model. Relative to the three-factor model, the biggest improvements in the number of non-zero intercepts are produced by the five-factor model when applied to the 32 *Size-OP-Inv* portfolios. As argued by Fama and French (2015), this is not surprising since these portfolios are formed on two variables (profitability and investment) not directly targeted by the three-factor model. Second, the best performance appears in the model of *SMB*, *HML*, and *RMW* or *CMA* with 32 *Size-B/M-OP* portfolios, as there are only one intercept left unexplained. However, this may not be a very robust result, since the model fares poorly in other portfolios, such as 32 *Size-B/M-Inv* portfolios and 32 *Size-OP-Inv* portfolios. Finally, similar to Fama and French (2015), we also find that *HML* is a redundant factor, as dropping *HML* from the five-factor model does not alter our results significantly.

Our proposed pleiotropy model could have practical applications. As stated, our method could determine how many and which intercepts in the time-series regressions in portfolio-level are statistically non-zero. For example, in table 8, we find that small stocks (index 1–5 in Panel A to Panel C, and index 1–8 in Panel D to Panel F) and big stocks (index 1–5 in Panel A to Panel C, and index 25–32 in Panel D to Panel F) can still hardly be explained well by existing factor models. On one hand, this suggests that further research is called for to incorporate the effect of small stocks; on the other hand, by providing potential directions for search in alpha, our finding may have implications for real-world investment decisions.

6. Conclusion and discussion

Factor models for asset pricing are widely used not only in academia but also in practice. The validity of the models relies on *GRS* test or other statistics to determine the explaining ability of those factors on excess returns of portfolios. In order to improve the interpretation and better judge between competing factor models, we recommend to use our proposed method of pleiotropy test, as the method could provide a more direct way to compare the validity of competing factor models and thus is an important and irreplaceable supplement to existing statistics in asset pricing models. The analytical results suggest that the widely used factor models are still imperfect interpretation of asset returns, as small stocks can still hardly be explained well by existing factor models, suggesting that further research is called for to incorporate the effect of small stocks.

At last, we close this work by pointing out that our basic idea can be extended in four directions: (1) Estimation

methods: In this paper, a known density function is assumed and maximum likelihood method is applied to estimation, but in practice, sometimes we do not know the density function, so maybe a more general estimation method, such as nonparametric estimation methods, can be applied to this problem in the future. (2) Complex and flexible models could be assumed, such as semiparametric models, functional time series models, high-dimensional models and so on. (3) Other factors in well-recognized research of asset pricing, such as Hou *et al.* (2015, 2020), Barillas and Shanken (2018), Stambaugh and Yuan (2017), Fama and French (2018), and Daniel *et al.* (2020) could also be considered. (4) When the distribution is misspecified, the sequential test based on quasi-likelihood seems to be a good solution, and our method could also be extended to this case with some resampling procedure to mimic the asymptotic null-distribution of the quasi-likelihood ratio test statistics.

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Data Availability statement

The data that support the findings of this study are openly available in Data Library of Kenneth R. French at <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/index.html>.

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Appendices

Appendix 1. Proofs of Lemmas and Theorems

A.1. Proof of Lemma 2.1

Proof We can proceed the proof in the same manner as Section 4.4 of Shao (2003). ■

A.2. Proof of Theorem 3.1

Proof The first assertion is shown in Schaid *et al.* (2016). Now we prove the second assertion. Expanding $l_n(\theta)$ around $\hat{\theta}_n$ by Taylor expansion, one can obtain that

$$l_n(\theta) = l_n(\hat{\theta}_n) - \frac{1}{2}(\theta - \hat{\theta}_n)^\top A_n(\theta - \hat{\theta}_n) + o_p\{(\theta - \hat{\theta}_n)^\top A_n(\theta - \hat{\theta}_n)\},$$

which implies that

$$\begin{aligned} & -2(l_n(\hat{\theta}_V) - l_n(\hat{\theta}_n)) \\ &= (\hat{\theta}_V - \hat{\theta}_n)^\top A_n(\hat{\theta}_V - \hat{\theta}_n) + o_p\{(\hat{\theta}_V - \hat{\theta}_n)^\top A_n(\hat{\theta}_V - \hat{\theta}_n)\} \\ &= \tilde{\theta}_V^\top A_n \tilde{\theta}_V + o_p(\tilde{\theta}_V^\top A_n \tilde{\theta}_V). \end{aligned}$$

Notice that $\tilde{\theta}_V = A_n^{-1} V^\top (V A_n^{-1} V^\top)^{-1} V \hat{\theta}_n$. Under the null hypothesis $V\theta_0 = 0$, we have

$$\begin{aligned} & \tilde{\theta}_V^\top A_n \tilde{\theta}_V \\ &= \hat{\theta}_n^\top V^\top (V A_n^{-1} V^\top)^{-1} V \hat{\theta}_n \\ &= \{\sqrt{n}(\hat{\theta}_n - \theta_0)\}^\top V^\top \{V(n^{-1} A_n)^{-1} V^\top\}^{-1} V \{\sqrt{n}(\hat{\theta}_n - \theta_0)\}. \end{aligned}$$

By Lemma 1, we have $n^{-1} A_n = I_n(\hat{\theta}_n)$ converges to $I(\theta_0)$ in probability and $V\{\sqrt{n}(\hat{\theta}_n - \theta_0)\} \xrightarrow{d} N(0, V\{I(\theta_0)\}^{-1} V^\top)$. Then by Slutsky's theorem, we have $\tilde{\theta}_V^\top A_n \tilde{\theta}_V = O_p(1)$. Thus

$$-2(l_n(\hat{\theta}_V) - l_n(\hat{\theta}_n)) = \tilde{\theta}_V^\top A_n \tilde{\theta}_V + o_p(1).$$

Therefore we complete the proof of Theorem 3.1. ■

A.3. Proof of Corollary 3.1

Proof (1). By Theorem 3.1, the null hypothesis $H_0 : V\theta_0 = 0$ yields that

$$\begin{aligned} \tilde{\theta}_V &= A_n^{-1} V^\top [V A_n^{-1} V^\top]^{-1} V \hat{\theta}_n \\ &= A_n^{-1} V^\top [V A_n^{-1} V^\top]^{-1} V [\theta_0 + n^{-1/2} I(\theta_0)^{-1/2} U(\theta_0) \\ &\quad + o_p(n^{-1/2})] \quad (\text{Lemma 2.1}) \\ &= A_n^{-1} V^\top [V A_n^{-1} V^\top]^{-1} V A_n^{-1/2} U(\theta_0) \\ &\quad + o_p(n^{-1/2}) \quad (\text{since } V\theta_0 = 0). \end{aligned}$$

Here $U(\theta_0) = n^{-1/2} \sum_{i=1}^n \{I(\theta_0)^{-1/2} \frac{\partial \log f(y_i | X_i^\top \beta, \gamma)}{\partial \theta} |_{\theta=\theta_0}\}$, and the last equality holds because $n^{-1} A_n \rightarrow I(\theta_0)$ in probability by the law of large number, which also gives that

$$\begin{aligned} \tilde{\theta}_V^\top A_n \tilde{\theta}_V &= \|A_n^{-1/2} V^\top [V A_n^{-1} V^\top]^{-1} V A_n^{-1/2} U(\theta_0) + o_p(n^{-1/2})\|^2 \\ &= \|P_V U(\theta_0)\|^2 + o_p(1). \end{aligned}$$

Since $P_V = A_n^{-1/2} V^\top [V A_n^{-1} V^\top]^{-1} V A_n^{-1/2}$ is an idempotent matrix of rank s and $U(\theta_0) \xrightarrow{d} N(0, I_{p+1})$, thus by central limit theorem and Slutsky's theorem, we can obtain that $\tilde{\theta}_V^\top A_n \tilde{\theta}_V \xrightarrow{d} \chi_s^2$.

(2). Following from Lemma 2.1, it holds that $\sqrt{n}(\hat{\theta}_n - \theta_0) = I(\theta_0)^{-1/2} U(\theta_0) + o_p(1)$, which yields that

$$\begin{aligned} \tilde{\theta}_V &= A_n^{-1} V^\top [V A_n^{-1} V^\top]^{-1} V \hat{\theta}_n \\ &= A_n^{-1} V^\top [V A_n^{-1} V^\top]^{-1} V [\theta_0 + n^{-1/2} I(\theta_0)^{-1/2} U(\theta_0) \\ &\quad + o_p(n^{-1/2})] \\ &= I(\theta_0)^{-1} V^\top [V I(\theta_0)^{-1} V^\top]^{-1} V \theta_0 + A_n^{-1/2} P_V U(\theta_0) \\ &\quad + o_p(n^{-1/2}). \end{aligned}$$

From the above proof of (1), it is clear that $\|P_V U(\theta_0)\|^2 \xrightarrow{d} \chi_s^2$, thus $\|P_V U(\theta_0)\|^2 = O_p(1)$. In addition, $\|A_n^{-1/2} \tilde{\theta}_V\|^2 = n \|I(\theta_0)^{-1/2} V^\top [V I(\theta_0)^{-1} V^\top]^{-1} V \theta_0 + o_p(1)\|^2 + O_p(1) + o_p(1)$. And since $\|I(\theta_0)^{-1/2} V^\top [V I(\theta_0)^{-1} V^\top]^{-1} V \theta_0\|^2 = O(1)$, we can know that $\tilde{\theta}_V^\top A_n \tilde{\theta}_V = O_p(n)$ if $V\theta_0 \neq 0$. ■

A.4. Proof of Theorem 3.2

Proof (1) If H_{00} holds, then $V_k \theta_0 = \mathbf{0}$ for $k = 0, 1, \dots, p$. Recall that $T_{nk} = \tilde{\theta}_{V_k}^\top A_n \tilde{\theta}_{V_k}$. By Corollary 3.1, we have $T_{n0} \xrightarrow{d} \chi_p^2$ and $T_{nk} \xrightarrow{d} \chi_{p-1}^2$ for $k = 1, \dots, p$, as $n \rightarrow \infty$.
 (2) If only one H_{0k} ($k = 1, \dots, p$) holds, which means $V_k \theta_0 = \mathbf{0}$ and $V_j \theta_0 \neq \mathbf{0}$ ($j = 0, 1, \dots, k-1, k+1, \dots, p$), then $T_{n0}, T_{n1}, \dots, T_{n,k-1}, T_{n,k+1}, \dots, T_{np} = O_p(n)$ and $T_{nk} \xrightarrow{d} \chi_{p-1}^2$. Recall that $T_{\min} = \min_{k=1, \dots, p} T_{nk}$. Notice that

$$P(T_{\min} > t) = P(\{T_{n1} > t\} \cap \dots \cap \{T_{np} > t\}) \\ \leq P(T_{nk} > t),$$

and

$$P(T_{\min} > t) = 1 - P(\{T_{n1} \leq t\} \cup \dots \cup \{T_{np} \leq t\}) \\ \geq 1 - \sum_{j=1}^p P(T_{nj} \leq t).$$

Based on the asymptotic properties for T_{nj} ($j = 1, \dots, p$), then we can obtain that

$$1 - F_{\chi_{p-1}^2}(t) \leq \lim_{n \rightarrow \infty} P(T_{\min} > t) \leq 1 - F_{\chi_{p-1}^2}(t)$$

Thus $T_{\min} \xrightarrow{d} \chi_{p-1}^2$ as $n \rightarrow \infty$.

(3) If all H_{0k} 's ($k = 0, \dots, p$) do not hold, which means $V_k \theta_0 \neq \mathbf{0}$ for any $k = 0, \dots, p$, then by Corollary 3.1, we have $T_{nk} = O_p(n)$ for $k = 0, \dots, p$.
 Hence, we complete the proof of Theorem 3.2. ■

Appendix 2. Relationship between proposed test statistics and GRS test statistics

Without loss of generality, we assume a same multivariate linear regression model as Gibbons *et al.* (1989):

$$y_{jt} = \alpha_j + \beta_j x_t + \varepsilon_{jt}, \quad \forall j = 1, \dots, p; \quad t = 1, \dots, n.$$

where y_{jt} is the excess return on asset j in period t and x_t represents the excess return on portfolio whose efficiency is being tested; and ε_{jt} is the disturbance term for asset j in period t . The disturbances are assumed to be jointly normally distributed each period with mean zero and nonsingular covariance matrix Σ , conditional on x . Also assume independence of the disturbances over time. In order to test the hypothesis $H_0: \alpha_j = 0, \forall j = 1, \dots, p$, Gibbons *et al.* (1989) proposed a 'Hotelling's T^2 test', while we propose a likelihood ratio test. Interestingly, there is some relationship between the two test statistics, thus we will investigate the derivation of GRS test statistic as a function of the our proposed test statistic or likelihood ratio criterion. Given the normality assumption, the likelihood function is

$$L(\alpha, \beta, \Sigma) = (2\pi)^{-\frac{pn}{2}} |\Sigma|^{-\frac{n}{2}} \exp \\ \times \left\{ -\frac{1}{2} \sum_{t=1}^n (\mathbf{y}_t - \alpha - \beta x_t)^\top \Sigma^{-1} (\mathbf{y}_t - \alpha - \beta x_t) \right\},$$

where $\mathbf{y}_t = (y_{1t}, \dots, y_{pt})^\top$, $\alpha = (\alpha_1, \dots, \alpha_p)^\top$, $\beta = (\beta_1, \dots, \beta_p)^\top$. Then the likelihood ratio criterion is

$$\lambda = \frac{\max_{\alpha, \beta, \Sigma} L(\beta, \Sigma)}{\max_{\alpha, \beta, \Sigma} L(\alpha, \beta, \Sigma)},$$

that is, the numerator is the maximum of the likelihood function for β, Σ in the parameter space restricted by the null hypothesis ($\alpha = 0$),

and the denominator is the maximum over the entire parameter space. When $\alpha = 0$, the likelihood function is maximized at

$$\hat{\beta}_w = \left(\frac{1}{n} \sum_{t=1}^n x_t^2 \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n x_t y_t \right), \\ \hat{\Sigma}_w = \frac{1}{n} \sum_{t=1}^n (\mathbf{y}_t - \hat{\beta}_w x_t)(\mathbf{y}_t - \hat{\beta}_w x_t)^\top.$$

When the parameters are unrestricted, the maximum occurs when α, β, Σ are defined by the maximum likelihood estimators of α, β and Σ ,

$$\hat{\alpha}_\Omega = \frac{1}{n} \sum_{t=1}^n \mathbf{y}_t - \frac{1}{n} \sum_{t=1}^n x_t \hat{\beta}, \\ \hat{\beta}_\Omega = \frac{n^{-1} \sum_{t=1}^n x_t \mathbf{y}_t - (n^{-1} \sum_{t=1}^n x_t)(n^{-1} \sum_{t=1}^n \mathbf{y}_t)}{n^{-1} \sum_{t=1}^n x_t^2 - (n^{-1} \sum_{t=1}^n x_t)^2}, \\ \hat{\Sigma}_\Omega = \frac{1}{n} \sum_{t=1}^n (\mathbf{y}_t - \hat{\alpha}_\Omega - \hat{\beta}_\Omega x_t)(\mathbf{y}_t - \hat{\alpha}_\Omega - \hat{\beta}_\Omega x_t)^\top.$$

Thus the likelihood ratio criterion is

$$\lambda = \frac{(2\pi)^{-pn/2} |\hat{\Sigma}_w|^{-n/2} e^{-pn/2}}{(2\pi)^{-pn/2} |\hat{\Sigma}_\Omega|^{-n/2} e^{-pn/2}} = \frac{|\hat{\Sigma}_\Omega|^{n/2}}{|\hat{\Sigma}_w|^{n/2}}.$$

By the proof of Lemma 8.3.1 in Anderson (2003), we have

$$n \hat{\Sigma}_w = n \hat{\Sigma}_\Omega + (\hat{\alpha}_\Omega - \alpha) A_{11.2} (\hat{\alpha}_\Omega - \alpha)^\top,$$

where $A = \begin{pmatrix} n & \sum_{t=1}^n x_t \\ \sum_{t=1}^n x_t & \sum_{t=1}^n x_t^2 \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$, $A_{11.2} = n - (\sum_{t=1}^n x_t)^2 / (\sum_{t=1}^n x_t^2) = n(1 + \hat{\theta}_p^2)^{-1}$. Since GRS test statistics $T_{GRS} = n(n-p-1) \hat{\alpha}_\Omega^\top \hat{\Sigma}^{-1} \hat{\alpha}_\Omega / \{p(n-2)(1 + \hat{\theta}_p^2)\}$, where $\hat{\Sigma} = n \hat{\Sigma}_\Omega / (n-2)$. Then under H_0 and by the Corollary A.3.1 of Anderson (2003), we have

$$\lambda^{2/n} = \frac{n |\hat{\Sigma}_\Omega|}{|n \hat{\Sigma}_\Omega + n \hat{\alpha}_\Omega \hat{\alpha}_\Omega^\top / (1 + \hat{\theta}_p^2)|} \\ = \frac{1}{1 + n(n-2)^{-1} \hat{\alpha}_\Omega^\top \hat{\Sigma}^{-1} \hat{\alpha}_\Omega / (1 + \hat{\theta}_p^2)} \\ = \frac{1}{1 + p T_{GRS} / (n-p-1)}.$$

On the other hand, our proposed test statistic $T_{n0} = -2 \log(\lambda)$. Thus we can conclude that

$$T_{GRS} = \frac{n-p-1}{p} \left\{ \exp(n^{-1} T_{n0}) - 1 \right\}.$$

Appendix 3. 1-order Vector Autoregression Model (VAR(1))

A.5. Models and notations

Consider the model

$$\mathbf{y}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \beta x_t + \mathbf{e}_t, \quad t = 2, \dots, T$$

where $\mathbf{y}_t$ and $\mathbf{y}_{t-1}$ are p -dimensional random vector, x_t is a scalar variable, $\mathbf{e}_t$ is white noise, and $\mathbf{e}_t \stackrel{i.i.d.}{\sim} N(\mathbf{0}, \Sigma)$. $\mathbf{c}$ is a constant, Φ and β are parameters of the model, and Φ is a $p \times p$ -dimensional

matrix, β is a p -dimensional vector. Classical estimation method for VAR(1) is by maximizing conditional log-likelihood function. It is easy to obtain the conditional likelihood function of $\mathbf{y}_2, \dots, \mathbf{y}_T$ on $\mathbf{y}_1$ as follows,

$$\begin{aligned} f(\mathbf{y}_2, \dots, \mathbf{y}_T | \mathbf{y}_1; \mathbf{c}, \Phi, \beta, \Sigma) \\ &= \prod_{t=2}^T f(\mathbf{y}_t | \mathbf{y}_{t-1}; \mathbf{c}, \Phi, \beta, \Sigma) \\ &= \prod_{t=2}^T (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp \\ &\quad \times \left\{ -\frac{1}{2} (\mathbf{y}_t - \mathbf{c} - \Phi \mathbf{y}_{t-1} - \beta x_t)^\top \Sigma^{-1} (\mathbf{y}_t - \mathbf{c} - \Phi \mathbf{y}_{t-1} - \beta x_t) \right\}. \end{aligned}$$

For notational simplicity, we denote $\Pi^\top = [\mathbf{c} \ \Phi \ \beta]$, and $X_t^\top = [1 \ \mathbf{y}_{t-1}^\top \ x_t]$. So the conditional log-likelihood function could be expressed as

$$\begin{aligned} l(\Pi, \Sigma) &= -\frac{p(T-1)}{2} \log(2\pi) - \frac{T-1}{2} \log |\Sigma| \\ &\quad - \frac{1}{2} \sum_{t=2}^T (\mathbf{y}_t - \Pi^\top X_t)^\top \Sigma^{-1} (\mathbf{y}_t - \Pi^\top X_t). \end{aligned}$$

The estimator of Π and Σ could be computed by solving the equations $\partial l(\Pi, \Sigma) / \partial \Pi = \mathbf{0}$ and $\partial l(\Pi, \Sigma) / \partial \Sigma = \mathbf{0}$. And the corresponding solutions are

$$\begin{aligned} \hat{\Pi} &= \left(\sum_{t=2}^T X_t X_t^\top \right)^{-1} \left(\sum_{t=2}^T X_t \mathbf{y}_t^\top \right), \\ \hat{\Sigma} &= \frac{1}{T-1} \sum_{t=2}^T (\mathbf{y}_t - \hat{\Pi}^\top X_t) (\mathbf{y}_t - \hat{\Pi}^\top X_t)^\top. \end{aligned}$$

Next, we need to compute the estimators under some constraints. In order to be consistent with the estimation method as before described, we define some new notations. Denote $\Phi = [\psi_1^\top \ \psi_2^\top \ \dots \ \psi_p^\top]^\top$, where ψ_j ($j = 1, \dots, p$) is the j th row vector of Φ . Then $\Pi^\top$ has another equivalent expression, that is, $\Pi^\top = \begin{pmatrix} c_1 & \psi_1 & \beta_1 \\ c_2 & \psi_2 & \beta_2 \\ \vdots & \vdots & \vdots \\ c_p & \psi_p & \beta_p \end{pmatrix}$. Let $\theta^\top = [c_1, \dots, c_p, \psi_1, \dots, \psi_p, \beta_1, \dots, \beta_p]$.

Thus parameter vector θ has $p^2 + 2p$ elements. In this way, the estimator $\hat{\theta}_T$ of θ could be obtained from some transformation process of $\hat{\Pi}$.

A.6. Test statistic for VAR(1)

Let $V_0 = [0_{p \times p} \ 0_{p \times p^2} \ I_{p \times p}]$, where $0_{p \times q}$ is a $p \times q$ -dimensional matrix in which all elements equal to 0, and $I_{p \times p}$ is an identity matrix. Therefore

$$V_0 \theta = 0 \iff \beta_1 = \beta_2 = \dots = \beta_p = 0.$$

When $k = 1, \dots, p$, we conduct the matrix V_k by deleting the k th row from V_0 (ignore β_k). In order to compute the test statistic $T_{Tk} = \hat{\theta}_T^\top V_k^\top [V_k A_T^{-1} V_k^\top]^{-1} V_k \hat{\theta}_T$, we need to obtain $A_T = -\sum_{t=2}^T \partial^2 \log f(\mathbf{y}_t | \hat{\theta}_T, \mathbf{y}_{t-1}) / \partial \theta \partial \theta^\top$. Denote $\varepsilon_k =$

$y_{tk} - c_k - \psi_k \mathbf{y}_{t-1} - \beta_k x_t$ and $\Omega = \Sigma^{-1}$. Then

$$\begin{aligned} \log f(\mathbf{y}_t | \theta, \mathbf{y}_{t-1}, \Sigma) \\ &= -\frac{p}{2} \log(2\pi) - \frac{1}{2} |\Sigma| \\ &\quad - \frac{1}{2} (\mathbf{y}_t - \mathbf{c} - \Phi \mathbf{y}_{t-1} - \beta x_t)^\top \Sigma^{-1} (\mathbf{y}_t - \mathbf{c} - \Phi \mathbf{y}_{t-1} - \beta x_t) \\ &= -\frac{p}{2} \log(2\pi) - \frac{1}{2} |\Sigma| - \frac{1}{2} \sum_{k=1}^p \sum_{j=1}^p \varepsilon_k \Omega_{kj} \varepsilon_j. \end{aligned}$$

Then we need to consider these first partial derivatives for each element, since

$$\begin{aligned} \frac{\partial \varepsilon_k \Omega_{kj} \varepsilon_j}{c_m} \\ &= \begin{cases} -2 \Omega_{mm} (y_{tm} - c_m - \psi_m \mathbf{y}_{t-1} - \beta_m x_t) & k = j = m, \\ -\Omega_{mj} (y_{tj} - c_j - \psi_j \mathbf{y}_{t-1} - \beta_j x_t) & k = m, j \neq m, \\ -\Omega_{km} (y_{tk} - c_k - \psi_k \mathbf{y}_{t-1} - \beta_k x_t) & k \neq m, j = m, \\ 0 & k \neq m, j \neq m, \end{cases} \end{aligned}$$

It is easy to obtain that

$$\frac{\partial \log f(\mathbf{y}_t | \theta, \mathbf{y}_{t-1}, \Sigma)}{\partial c_m} = \sum_{k=1}^p \Omega_{mk} (y_{tk} - c_k - \psi_k \mathbf{y}_{t-1} - \beta_k x_t).$$

Thus some second partial derivatives are as follows

$$\begin{aligned} \frac{\partial^2 \log f(\mathbf{y}_t | \theta, \mathbf{y}_{t-1}, \Sigma)}{\partial c_m \partial c_n} &= -\Omega_{mn}, \\ \frac{\partial^2 \log f(\mathbf{y}_t | \theta, \mathbf{y}_{t-1}, \Sigma)}{\partial c_m \partial \psi_n} &= -\Omega_{mn} \mathbf{y}_{t-1}^\top, \\ \frac{\partial^2 \log f(\mathbf{y}_t | \theta, \mathbf{y}_{t-1}, \Sigma)}{\partial c_m \partial \beta_n} &= -\Omega_{mn} x_t. \end{aligned}$$

Similar computation can also be conducted.

$$\begin{aligned} \frac{\partial^2 \log f(\mathbf{y}_t | \theta, \mathbf{y}_{t-1}, \Sigma)}{\partial \theta \partial \theta^\top} \\ &= \begin{bmatrix} \Omega & \Omega \otimes \mathbf{y}_{t-1}^\top & \Omega x_t \\ \Omega \otimes \mathbf{y}_{t-1} & \Omega \otimes \mathbf{y}_{t-1} \mathbf{y}_{t-1}^\top & x_t \Omega \otimes \mathbf{y}_{t-1} \\ x_t \Omega & x_t \Omega \otimes \mathbf{y}_{t-1}^\top & x_t^2 \Omega \end{bmatrix} \\ &= \begin{bmatrix} \Sigma^{-1} & \Sigma^{-1} \otimes \mathbf{y}_{t-1}^\top & \Sigma^{-1} x_t \\ \Sigma^{-1} \otimes \mathbf{y}_{t-1} & \Sigma^{-1} \otimes \mathbf{y}_{t-1} \mathbf{y}_{t-1}^\top & x_t \Sigma^{-1} \otimes \mathbf{y}_{t-1} \\ x_t \Sigma^{-1} & x_t \Sigma^{-1} \otimes \mathbf{y}_{t-1}^\top & x_t^2 \Sigma^{-1} \end{bmatrix}. \end{aligned}$$

Thus, corresponding estimators for A_T and T_{Tk} can be easily obtained.

A.7. Simulation study for VAR(1)

In the model of VAR(1), we consider two tests. One is testing for pleiotropy and the other is a sequential testing for the accuracy. The model is as follows

$$\mathbf{y}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \beta x_t + \varepsilon_t, \quad t = 2, \dots, T \quad (\text{A1})$$

where $\mathbf{y}_t$ and $\mathbf{y}_{t-1}$ are p -dimensional random vectors, x_t is a scalar variable, ε_t is white noise, and $\varepsilon_t \stackrel{i.i.d.}{\sim} N(\mathbf{0}, \Sigma)$. $\mathbf{c}$ is a constant, Φ and β are parameters of the model, and Φ is a $p \times p$ -dimensional matrix, β is a p -dimensional vector. Let the intercept equals to 1, i.e. $\beta_0 = 1$ and the coefficient matrix Φ for $\mathbf{y}_{t-1}$ is an identity matrix I_p .

Scenario 1: Testing for pleiotropy

For the null hypothesis with no pleiotropy, we perform two sets of simulations. (i) all $\beta_j = 0$ ($j = 1, \dots, p$); (ii) $\beta_1 = 1$ and all

Table A1. Type I error of T_{n0} and $T_{\min}$ for VAR(1) under H_0 .

n	p	ρ_ε	$H_{00} : \beta_1 = \dots = \beta_p = 0$				$H_{01} : \beta_1 = 1, \beta_j = 0 (j \neq 1)$			
			0.05		0.01		0.05		0.01	
			T_{n0}	$T_{\min}$	T_{n0}	$T_{\min}$	T_{n0}	$T_{\min}$	T_{n0}	$T_{\min}$
200	4	0	0.0657	0.0049	0.0137	0.0003	1	0.0637	1	0.0151
		0.2	0.0606	0.0043	0.0162	0.0005	1	0.0617	1	0.0140
		0.5	0.0671	0.0068	0.0180	0.0005	1	0.0638	1	0.0164
		0.8	0.0647	0.0096	0.0156	0.0017	1	0.0666	1	0.0160
	6	0	0.0729	0.0052	0.0190	0.0003	1	0.0717	1	0.0199
		0.2	0.0741	0.0063	0.0193	0.0008	1	0.0722	1	0.0181
		0.5	0.0702	0.0065	0.0161	0.0006	1	0.0766	1	0.0182
		0.8	0.0754	0.0107	0.0200	0.0020	1	0.0739	1	0.0195
	10	0	0.0973	0.0114	0.0252	0.0018	1	0.1034	1	0.0303
		0.2	0.1005	0.0139	0.0305	0.0018	1	0.1041	1	0.0310
		0.5	0.1010	0.0138	0.0270	0.0023	1	0.1059	1	0.0324
		0.8	0.0985	0.0135	0.0313	0.0021	1	0.1170	1	0.0344
500	4	0	0.0550	0.0040	0.0128	0.0002	1	0.0576	1	0.0139
		0.2	0.0553	0.0047	0.0115	0.0000	1	0.0587	1	0.0112
		0.5	0.0587	0.0082	0.0130	0.0007	1	0.0522	1	0.0103
		0.8	0.0508	0.0066	0.0113	0.0010	1	0.0563	1	0.0121
	6	0	0.0612	0.0049	0.0140	0.0003	1	0.0549	1	0.0113
		0.2	0.0597	0.0054	0.0137	0.0010	1	0.0567	1	0.0119
		0.5	0.0598	0.0061	0.0131	0.0006	1	0.0603	1	0.0143
		0.8	0.0593	0.0082	0.0144	0.0012	1	0.0595	1	0.0121
	10	0	0.0693	0.0064	0.0165	0.0007	1	0.0665	1	0.0152
		0.2	0.0673	0.0083	0.0149	0.0007	1	0.0698	1	0.0174
		0.5	0.0654	0.0090	0.0159	0.0014	1	0.0689	1	0.0168
		0.8	0.0658	0.0072	0.0142	0.0008	1	0.0679	1	0.0154

Notes: T_{n0} tests the null hypothesis that all β_j 's are zero, and $T_{\min}$ tests the null hypothesis that there is only one nonzero β_j .

other $\beta_j = 0$ ($j = 2, \dots, p$). The significance level is set 0.05 or 0.01. And we generate x from a univariate standard normal distribution and the random errors from a multivariate normal distribution with a covariance of exchangeable correlation structure (to be $\rho_\varepsilon = 0, 0.2, 0.5, 0.8$) and variance of 1. Consider three different dimensions of y_t , that is, $p = 4, 6$, or 10 , and three different sample sizes $n = 200, 500$. To evaluate the power of our proposed method for pleiotropy, we set $p = 6$ or 10 , and the number of nonzero β_j 's is 3 or 5. And the marginal effect of these nonzero β_j 's is 0.1, 0.15 and 0.2. The sample size is assumed as $n = 500$ for the study of power. Other settings, such as the distribution of random errors or covariate, are as before. All simulations are repeated 10 000 times.

Tables A1–A2 present type I error rates and power for T_{n0} and $T_{\min}$ based on our proposed method. The results when all β_j 's equal zero, $T_{\min}$ tends to be very conservative (table A1). This is reasonable, since under H_{00} , the statistic $T_{\min}$ is based on the minimum of p correlated χ^2_{p-1} statistics. It is not easy to find the asymptotic distribution of $T_{\min}$ under H_{00} . But that has little effect on the test for H_{00} , because the type I error of T_{n0} performs well which is close to the nominal level. So in this case, T_{n0} plays a major role. table A1 illustrates that the type I error rates of our proposed method are near with the significance level with slight fluctuation, And in

table A2, it is obvious to find that power increases with more nonzero β_j 's, increasing correlation, and higher marginal effect of coefficients which are nonzero. Generally speaking, the procedure has well-controlled type I error rates and comparable power for testing the pleiotropy.

Scenario 2: Sequential testing for the accuracy of our proposed method

We conduct this simulation to evaluate the accuracy of choosing the correct number of nonzero β_j 's. For example, if $p = 6$, let the true number of nonzero β_j 's varies from 1 to 6, then we record the estimated numbers from 10 000 simulations of our proposed method, and compute the frequency. Assume that $p = 6$ or 10 , the marginal effect of coefficients is 1, the correlation $\rho_\varepsilon = 0$, and the sample size $n = 500$. The significance level α is 0.05 or 0.01.

The results in tables A3–A4 show that the accuracy rates are near to $1 - \alpha$, which implies that the sequential testing procedure is powerful. An interesting thing is that most false estimators are overestimated. This is consistent with table A1. Fortunately, the rate could be controlled by the significance level. In general case, sequential testing procedure could select the number of nonzero coefficients relatively effectively and correctly.

Table A2. Empirical power of T_{n0} and $T_{\min}$ for VAR(1) when sample size $n = 500$.

No. nonzero β_j 's	β_j	ρ_ε	$p = 6$				$p = 10$			
			0.05		0.01		0.05		0.01	
			T_{n0}	$T_{\min}$	T_{n0}	$T_{\min}$	T_{n0}	$T_{\min}$	T_{n0}	$T_{\min}$
3	0.10	0	0.8483	0.4471	0.6667	0.2100	0.7576	0.3424	0.5509	0.1372
		0.2	0.7777	0.4284	0.5718	0.1987	0.7488	0.3753	0.5421	0.1674
		0.5	0.8856	0.6089	0.7335	0.3618	0.9174	0.6402	0.7930	0.3845
		0.8	0.9988	0.9842	0.9942	0.9358	0.9998	0.9933	0.9988	0.9696
	0.15	0	0.9964	0.9268	0.9839	0.7885	0.9906	0.8560	0.9637	0.6599
		0.2	0.9929	0.9175	0.9665	0.7742	0.9909	0.8852	0.9608	0.7051
		0.5	0.9990	0.9836	0.9942	0.9306	0.9994	0.9850	0.9978	0.9466
		0.8	1	1	1	0.9999	1	1	1	1
	0.20	0	0.9999	0.9985	0.9998	0.9907	1	0.9950	0.9997	0.9756
		0.2	0.9999	0.9985	0.9998	0.9900	1	0.9965	0.9997	0.9820
		0.5	1	1	1	0.9994	1	1	1	0.9996
		0.8	1	1	1	1	1	1	1	1
5	0.10	0	0.9807	0.8646	0.9265	0.6710	0.9571	0.7636	0.8662	0.5363
		0.2	0.8605	0.6555	0.6879	0.4145	0.8952	0.6680	0.7410	0.4242
		0.5	0.8142	0.5186	0.6299	0.2847	0.9706	0.8593	0.9001	0.6761
		0.8	0.9753	0.4324	0.9203	0.2196	1	0.9996	0.9997	0.9983
	0.15	0	1	0.9996	0.9999	0.9965	0.9998	0.9973	0.9996	0.9861
		0.2	0.9975	0.9877	0.9893	0.9462	0.9993	0.9916	0.9953	0.9620
		0.5	0.9953	0.9319	0.9790	0.8079	0.9999	0.9993	0.9996	0.9964
		0.8	1	0.8159	0.9998	0.6151	1	1	1	1
	0.20	0	1	1	1	1	1	1	1	1
		0.2	1	1	1	0.9997	1	1	1	0.9999
		0.5	1	0.9979	0.9997	0.9871	1	1	1	1
		0.8	1	0.9772	1	0.9244	1	1	1	1

Notes: T_{n0} tests the null hypothesis that all β_j 's are zero, and $T_{\min}$ tests the null hypothesis that there is only one nonzero β_j .

Table A3. Sequential testing for the accuracy of identifying the true number of nonzero β_j 's when $n = 500$ and $p = 6$.

True No.	Est.1	Est.2	Est.3	Est.4	Est.5	Est.6
Estimated number of nonzero β_j 's (significance level 0.05)						
True.1	0.9392	0.0567	0.0037	0.0003	0.0001	0
True.2	0	0.9450	0.0521	0.0027	0.0002	0
True.3	0	0	0.9441	0.0527	0.0031	0.0001
True.4	0	0	0	0.9426	0.0541	0.0033
True.5	0	0	0	0	0.9380	0.0620
True.6	0	0	0	0	0	1
Estimated number of nonzero β_j 's (significance level 0.01)						
True.1	0.9868	0.0129	0.0003	0	0	0
True.2	0	0.9856	0.0141	0.0003	0	0
True.3	0	0	0.9876	0.0123	0.0001	0
True.4	0	0	0	0.9879	0.0120	0.0001
True.5	0	0	0	0	0.9874	0.0126
True.6	0	0	0	0	0	1

Appendix 4. QQ-plots for the data analysis

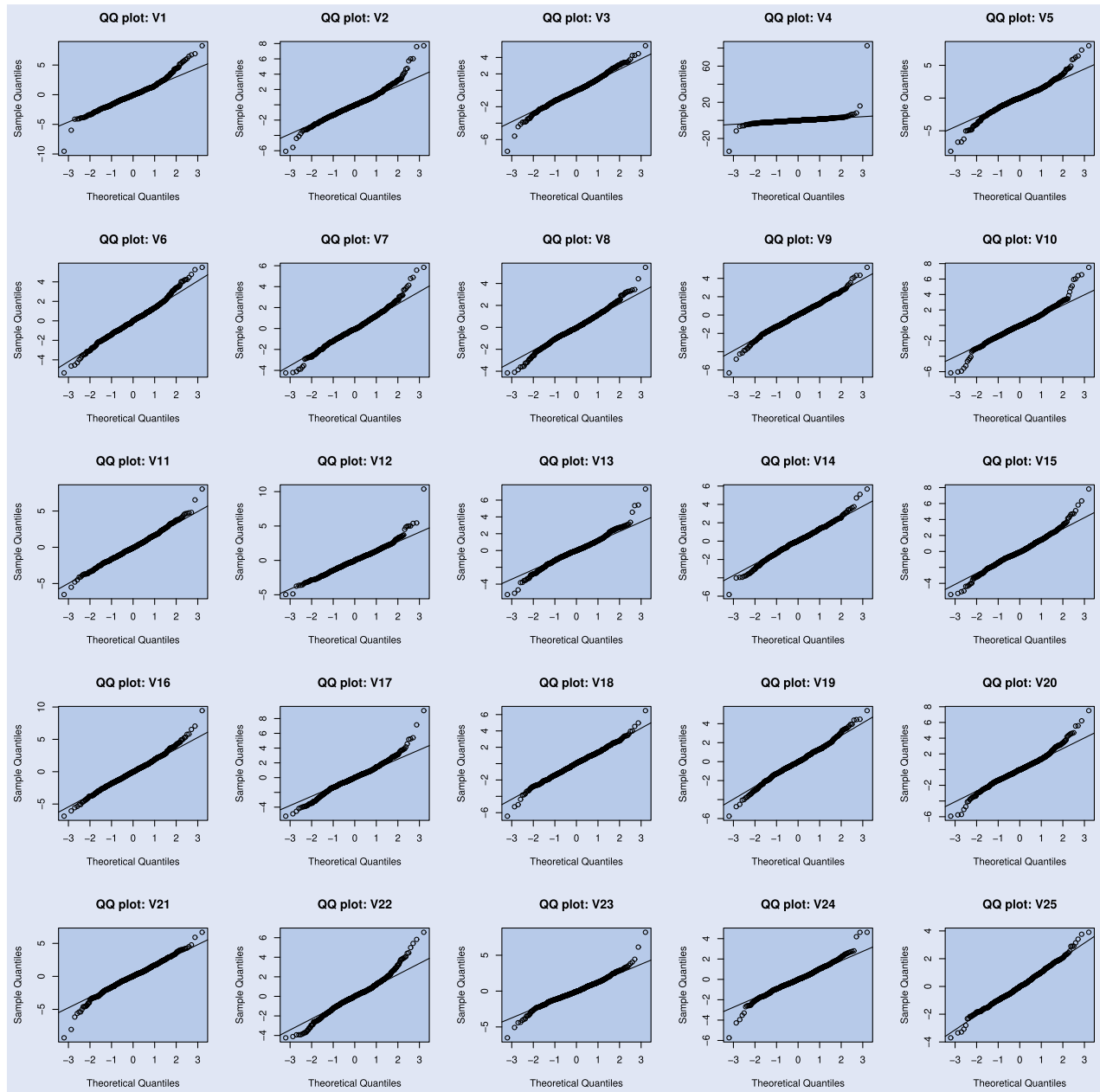


Figure A1. QQ-plots for 25 Size-OP portfolios.

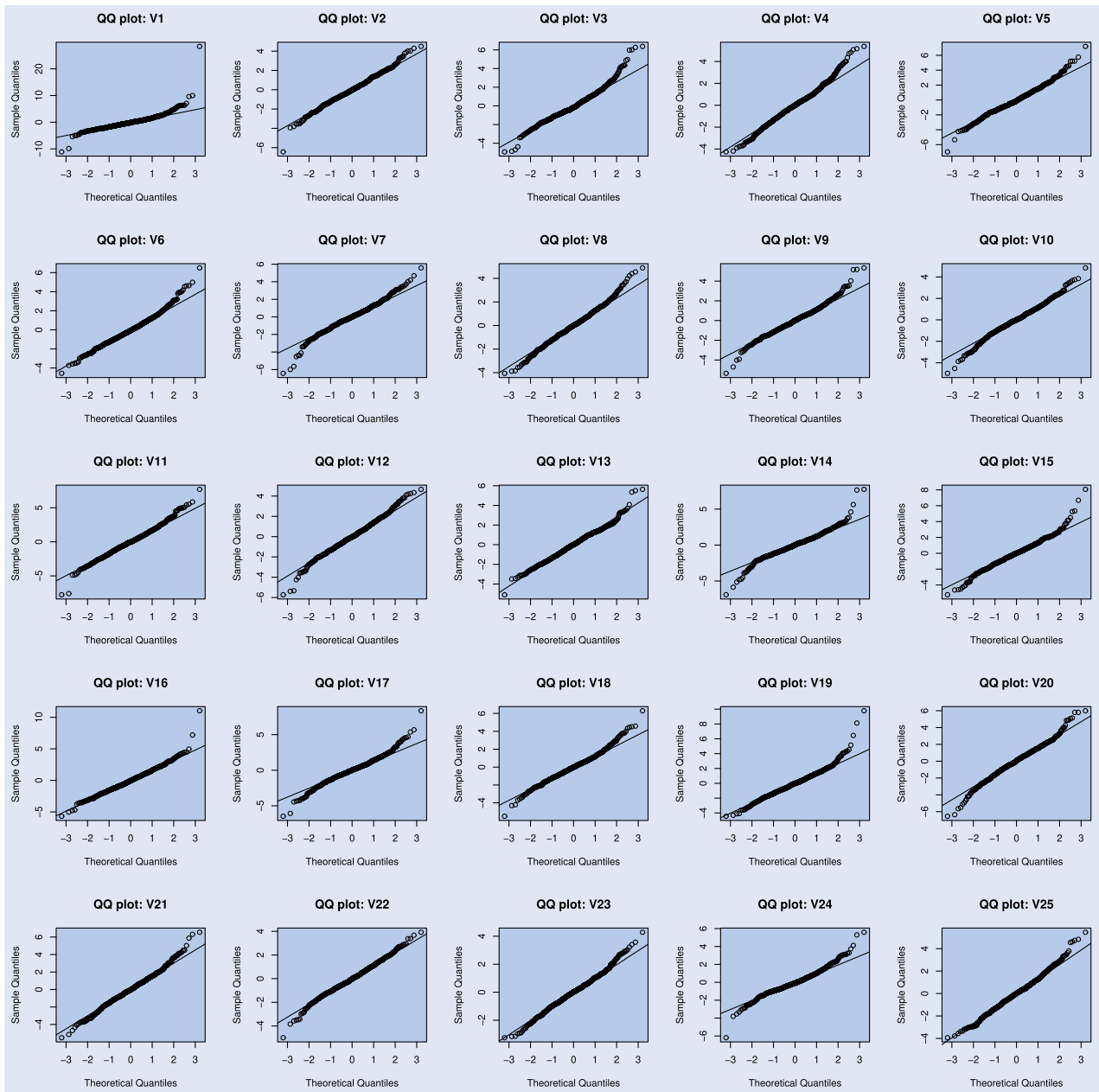


Figure A2. QQ-plots for 25 Size-Inv portfolios.



Figure A3. QQ-plots for 32 Size-B/M-OP portfolios.

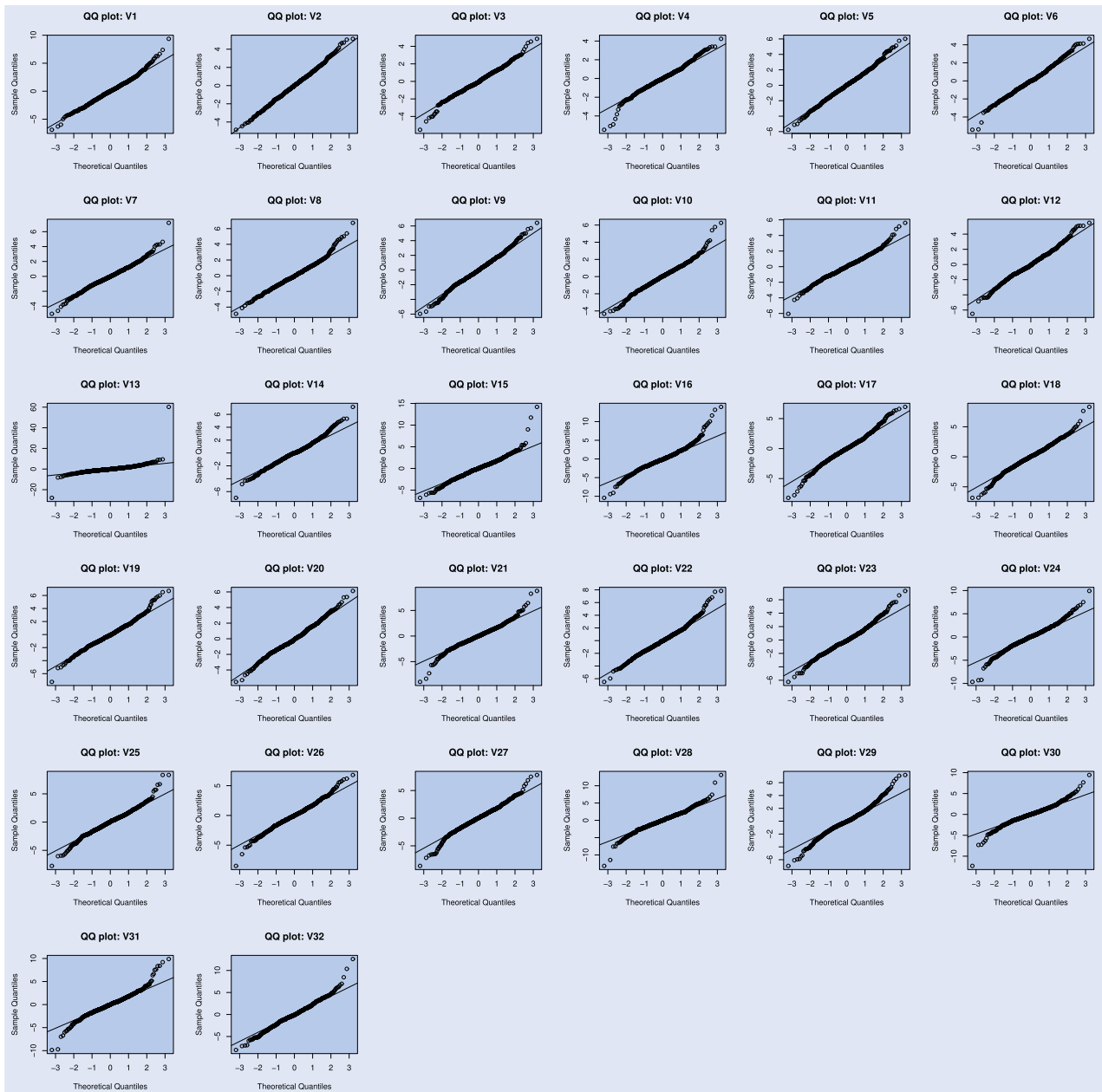


Figure A4. QQ-plots for 32 Size-B/M-Inv portfolios.

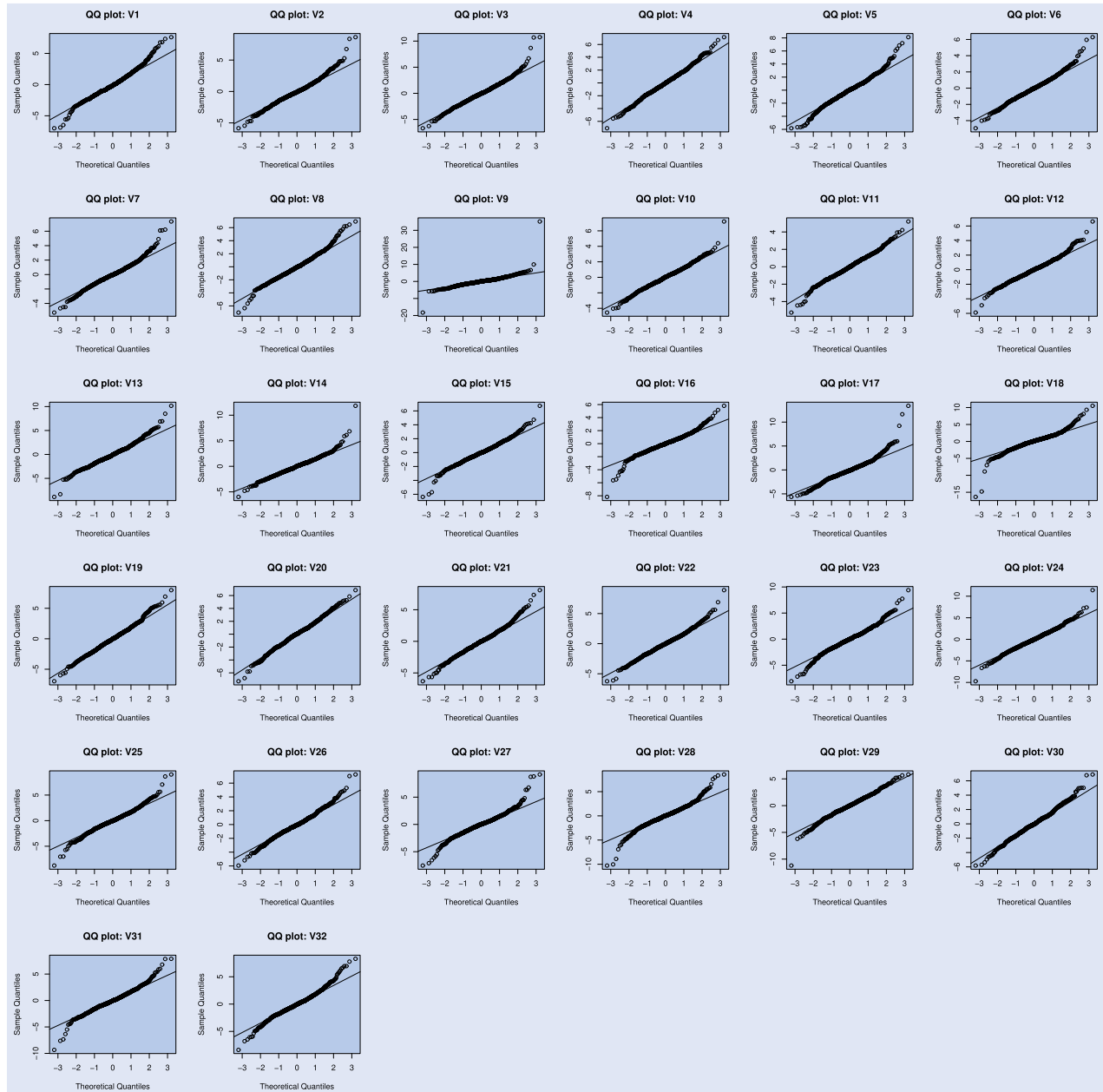


Figure A5. QQ-plots for 32 Size-OP-Inv portfolios.